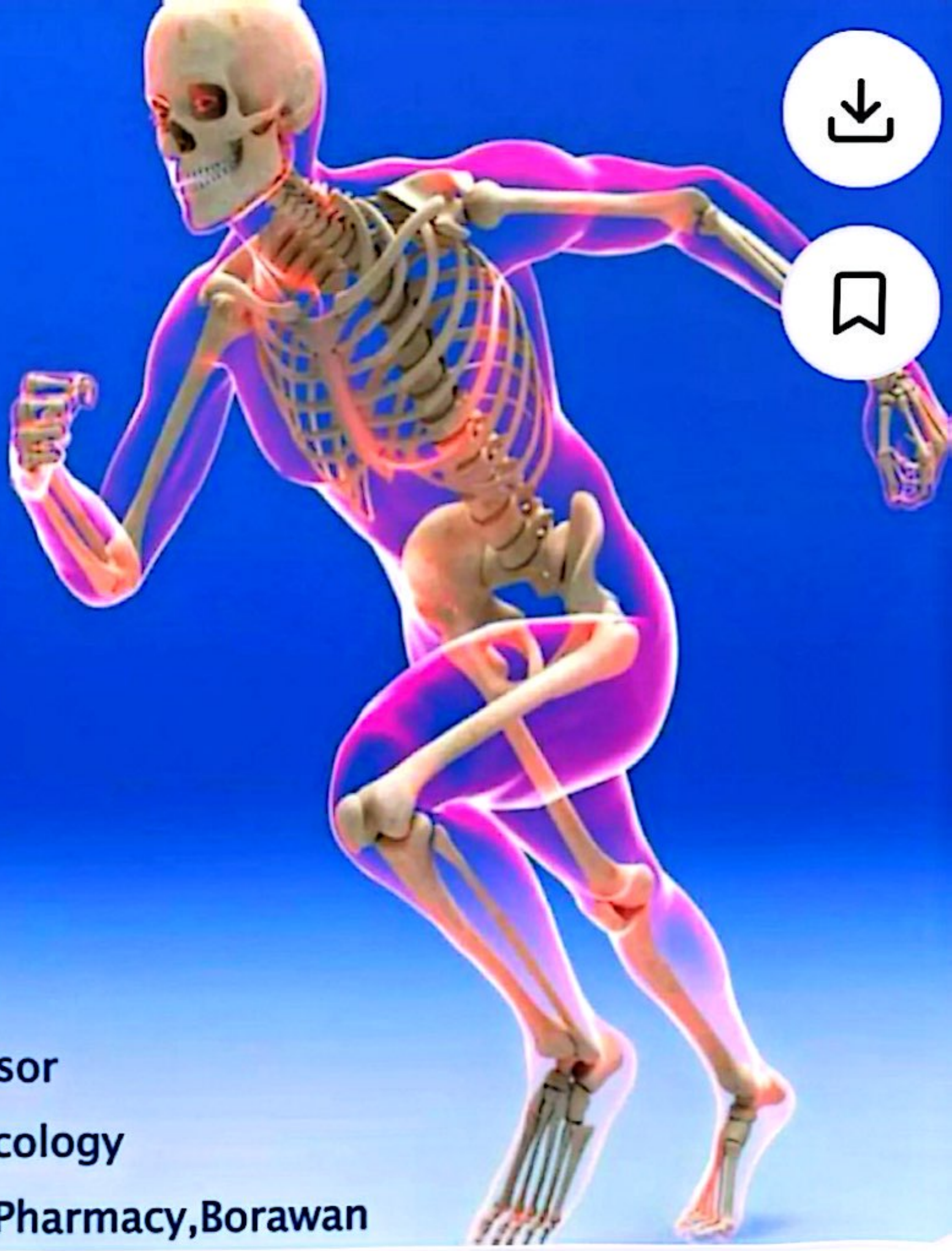
Skeletal System



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Associate Professor
Dept. of Pharmacology
GRY Institute of Pharmacy,Borawan

SKELETAL SYSTEM

- COMPOSED OF:
  - –Bones
  - –Cartilage
  - –Joints
  - –Ligaments

Functions of Skeletal System

- **SUPPORT**: Hard framework that supports and anchors the soft organs of the body.
- **PROTECTION**: Surrounds organs such as the brain and spinal cord.
- **MOVEMENT**: Allows for muscle attachment therefore the bones are used as levers.
- **STORAGE**: Minerals and lipids are stored within bone material.
- **BLOOD CELL FORMATION**: The bone marrow is responsible for blood cell production.

Bone Markings

- Bone Surface is not smooth, but shows:
- Bone markings which reveal where:

-muscles, tendons, and ligaments attatched

-nerves and blood vessels pass

*bone marking may be:

1-projections or processes or

2-depressions or cavities

Structure

- Compact bone
  - Outer layer of bone, very hard and dense.
  - Organized in structural units called Haversian systems.
  - Matrix is composed of Ca salts (Ca carbonate and Ca phosphate)
  - Osteocytes – living bone cells that live in matrix.
- Porous (Spongy) bone
  - Located in the ends of long bones.
  - Many spaces that are filled with red bone marrow which produces bone cells.

Structure

- Spongy bone
  - Trabeculae – needle-like threads of spongy bone that surround the spaces. Add strength to this portion of the bone.
- Cartilage
  - Matrix is a firm gel with chondrocytes suspended in the matrix.

Classification of Bones

- Long bones
  - Typically longer than wide
  - Have a shaft with heads at both ends
  - Contain mostly compact bone
    - Examples: Femur, humerus

Classification of Bones

- Short bones
  - Generally cube-shape
  - Contain mostly spongy bone
    - Examples: Carpals, tarsals

Classification of Bones on the Basis of Shape

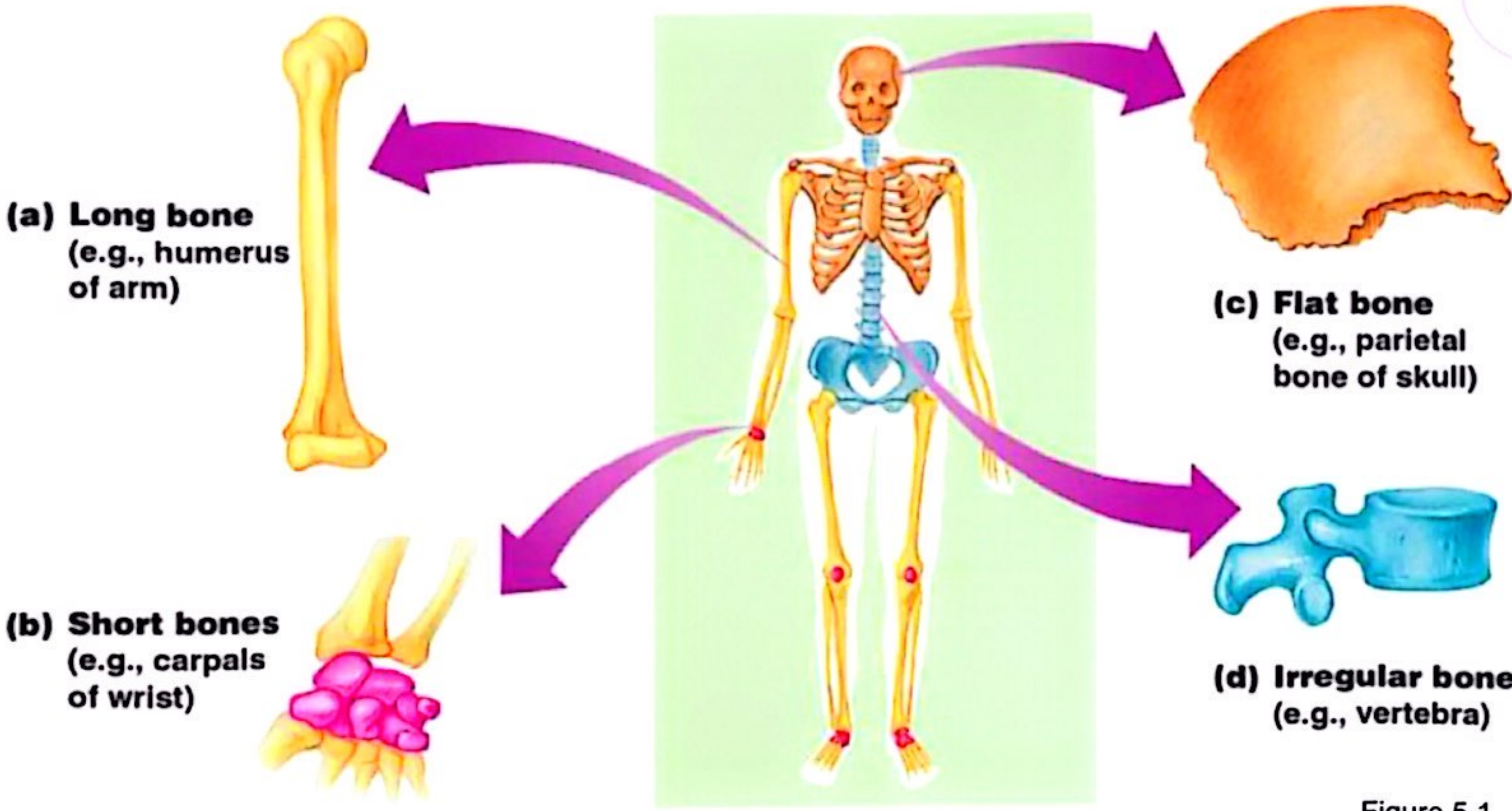


Figure 5.1

Classification of Bones

- Flat bones
  - Thin and flattened
  - Usually curved
  - Thin layers of compact bone around a layer of spongy bone
    - Examples: Skull, ribs, sternum

Classification of Bones

- Irregular bones
  - Irregular shape
  - Do not fit into other bone classification categories
    - Example: Vertebrae and hip

Slide 5.5b

Gross Anatomy of a Long Bone

- Diaphysis
  - Shaft
  - Composed of compact bone
- Epiphysis
  - Ends of the bone
  - Composed mostly of spongy bone

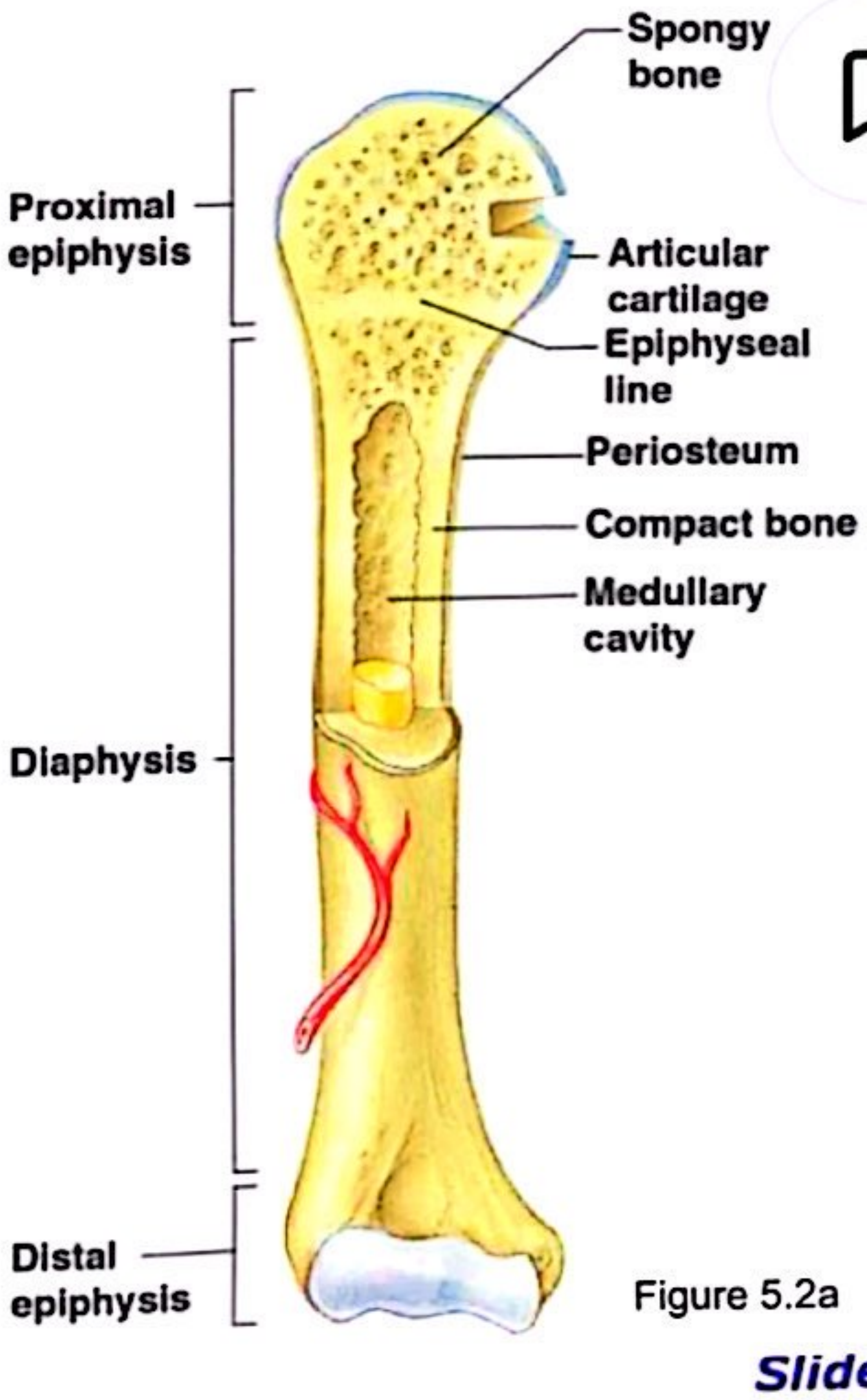


Figure 5.2a

Slide 5.6

Structures of a Long Bone

- Periosteum
  - Outside covering of the diaphysis
  - Fibrous connective tissue membrane
- Sharpey's fibers
  - Secure periosteum and underlying bone
- Arteries
  - Supply bone cells with nutrients

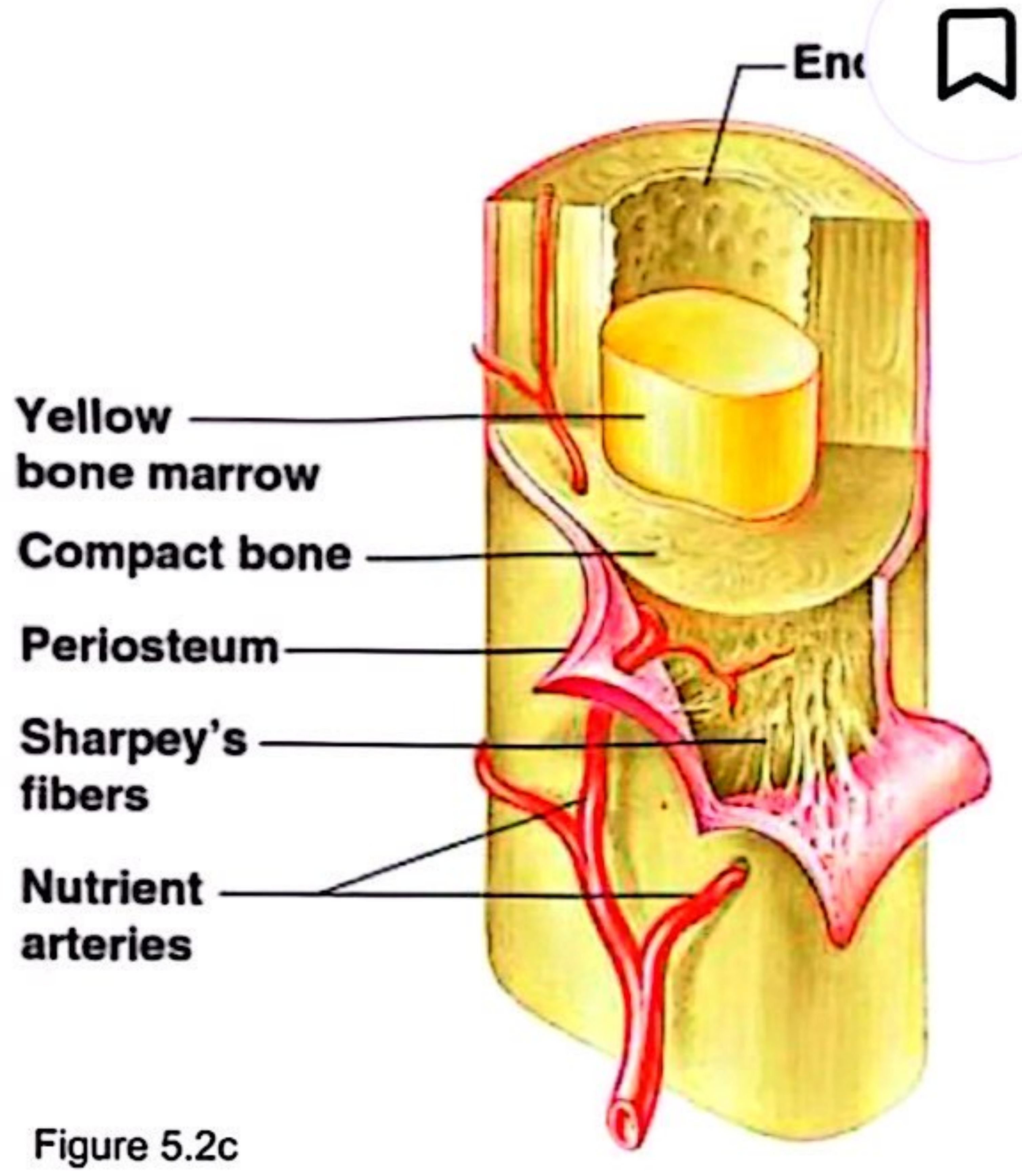


Figure 5.2c

Structures of a Long Bone

- Articular cartilage
  - Covers the external surface of the epiphyses
  - Made of hyaline cartilage
  - Decreases friction at joint surfaces

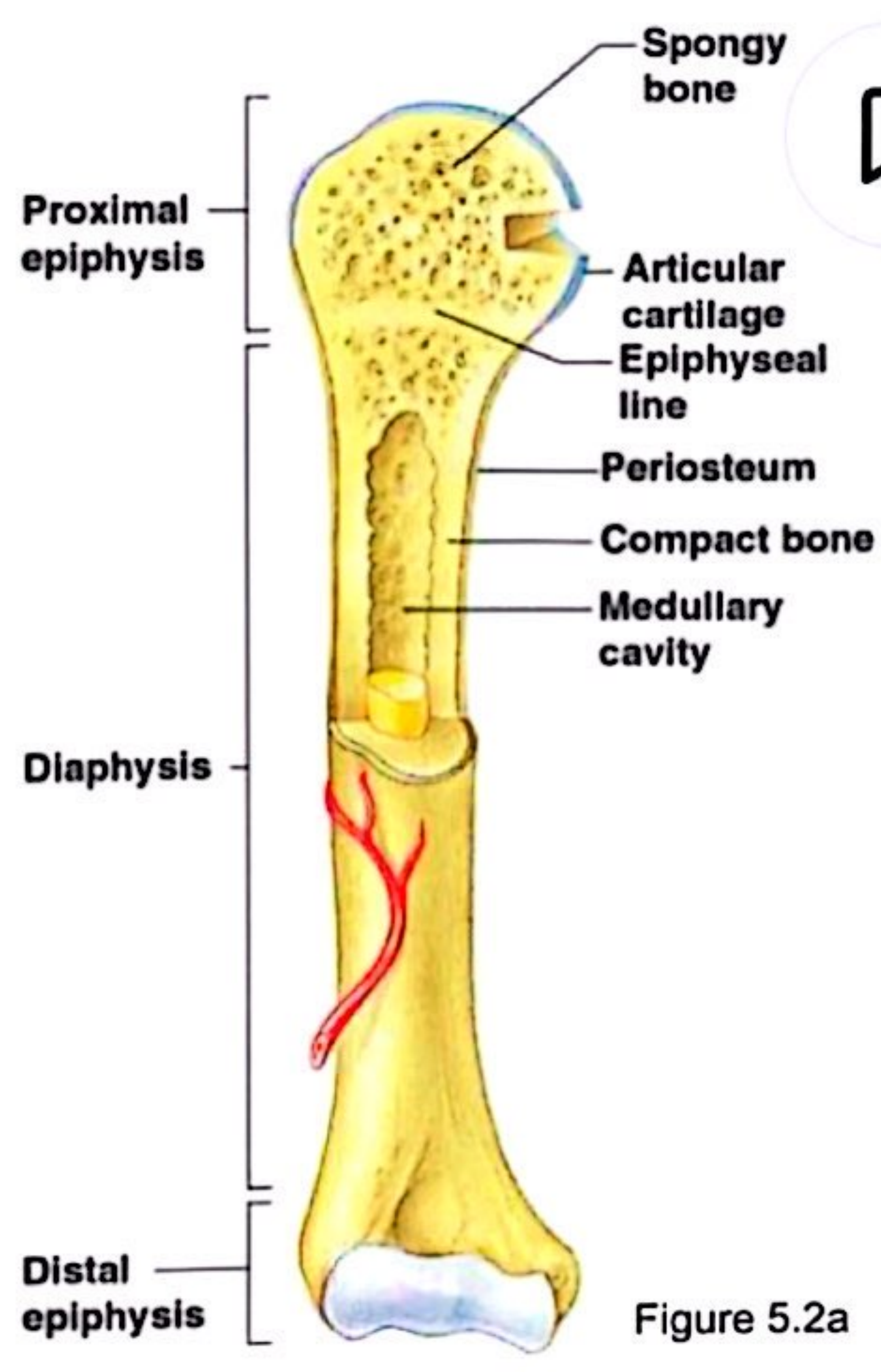


Figure 5.2a

Structures of a Long Bone

- Articular cartilage
  - Covers the external surface of the epiphyses
  - Made of hyaline cartilage
  - Decreases friction at joint surfaces

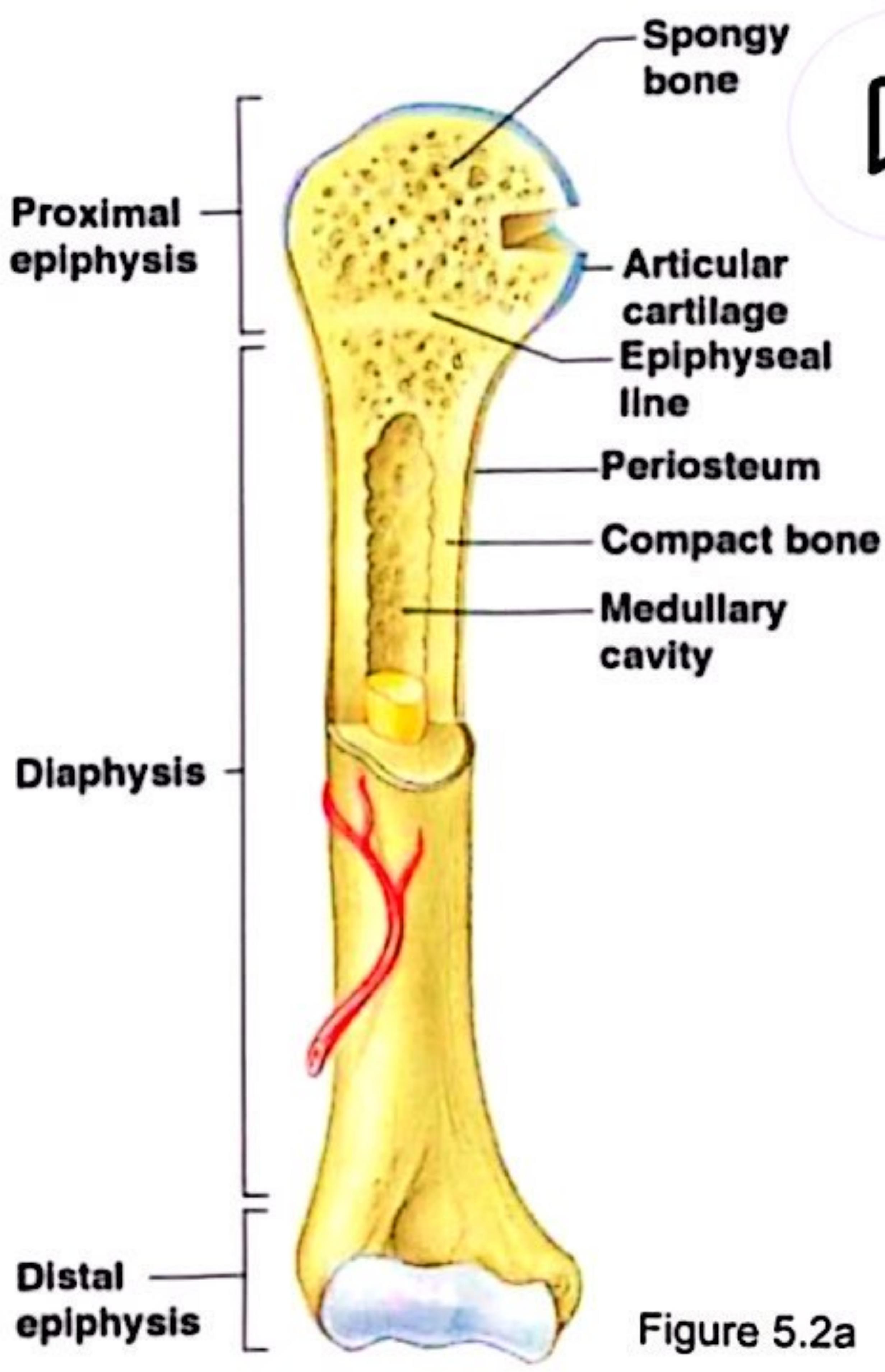


Figure 5.2a

Slide 5.8a

Structures of a Long Bone

- Medullary cavity
  - Cavity of the shaft
  - Contains yellow marrow (mostly fat) in adults
  - Contains red marrow (for blood cell formation)

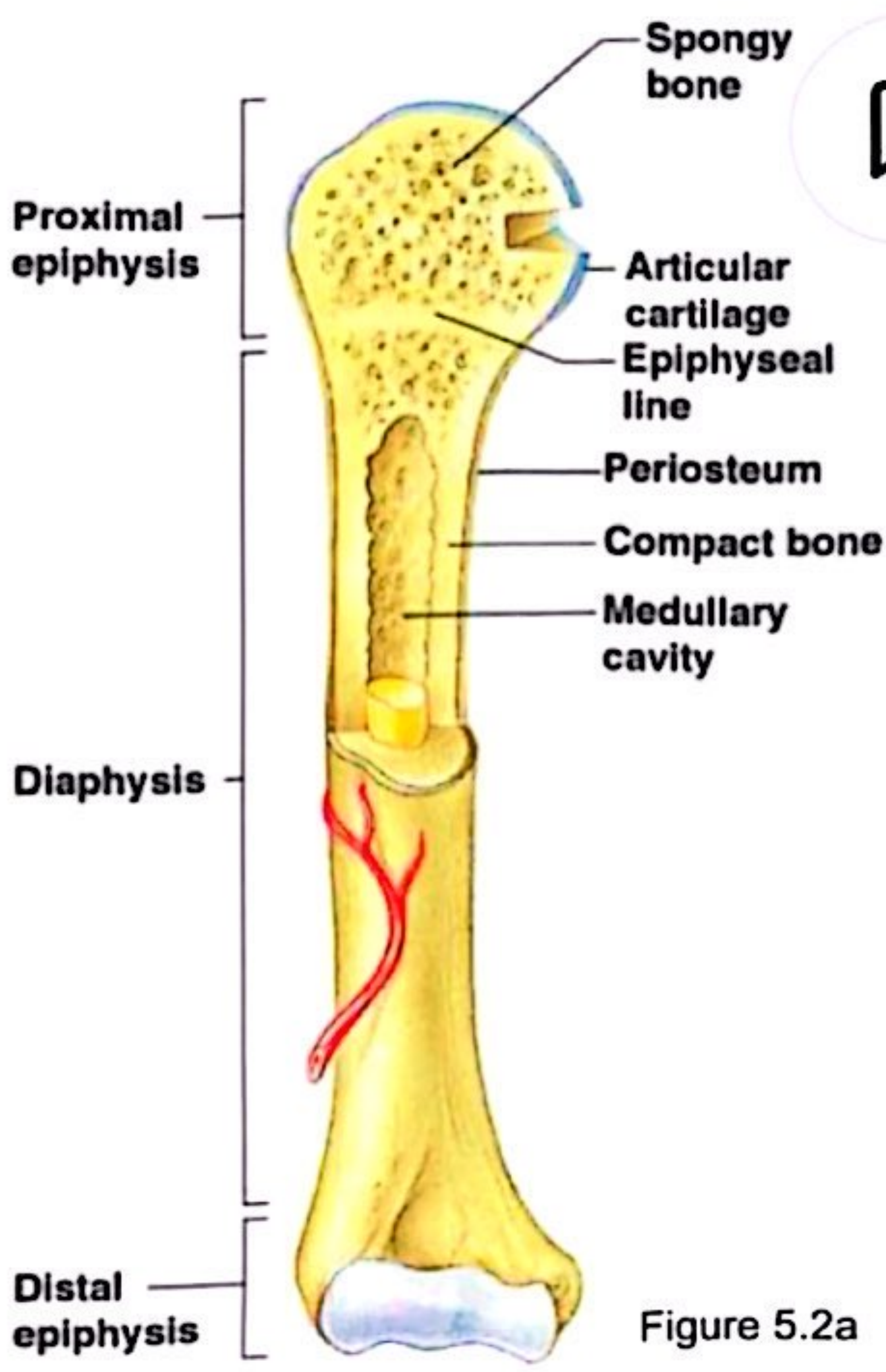


Figure 5.2a

Slide 5.8b

Microscopic Anatomy of Bone

- Osteon (Haversian System)
  - A unit of bone
- Central (Haversian) canal
  - Opening in the center of an osteon
  - Carries blood vessels and nerves
- Perforating (Volkman's) canal
  - Canal perpendicular to the central canal
  - Carries blood vessels and nerves

Slide 5.10a

Microscopic Anatomy of Bone

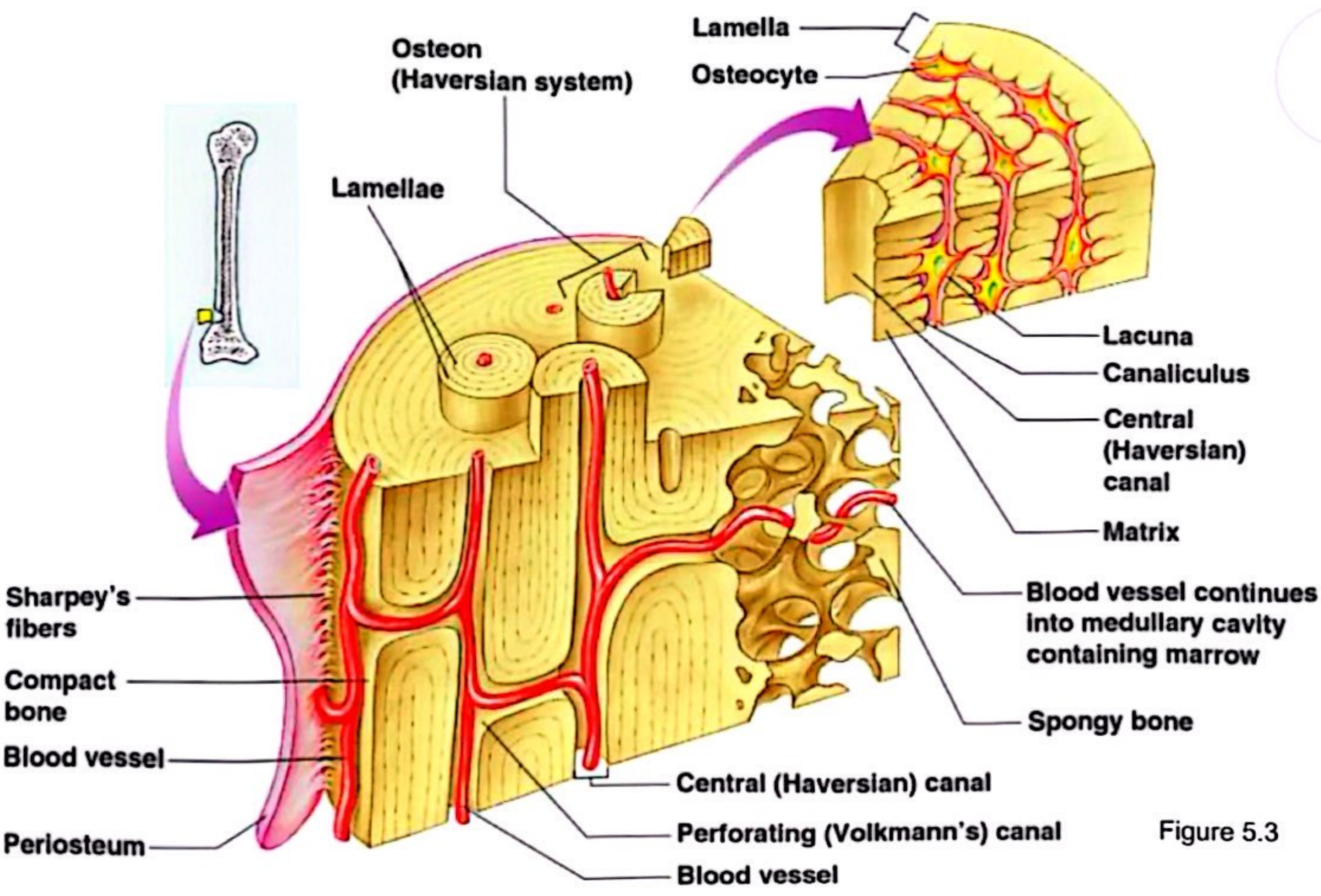


Figure 5.3

Slide 5.10b

Microscopic Anatomy of Bone

- Lacunae
  - Cavities containing bone cells (osteocytes)
  - Arranged in concentric rings
- Lamellae
  - Rings around the central canal
  - Sites of lacunae

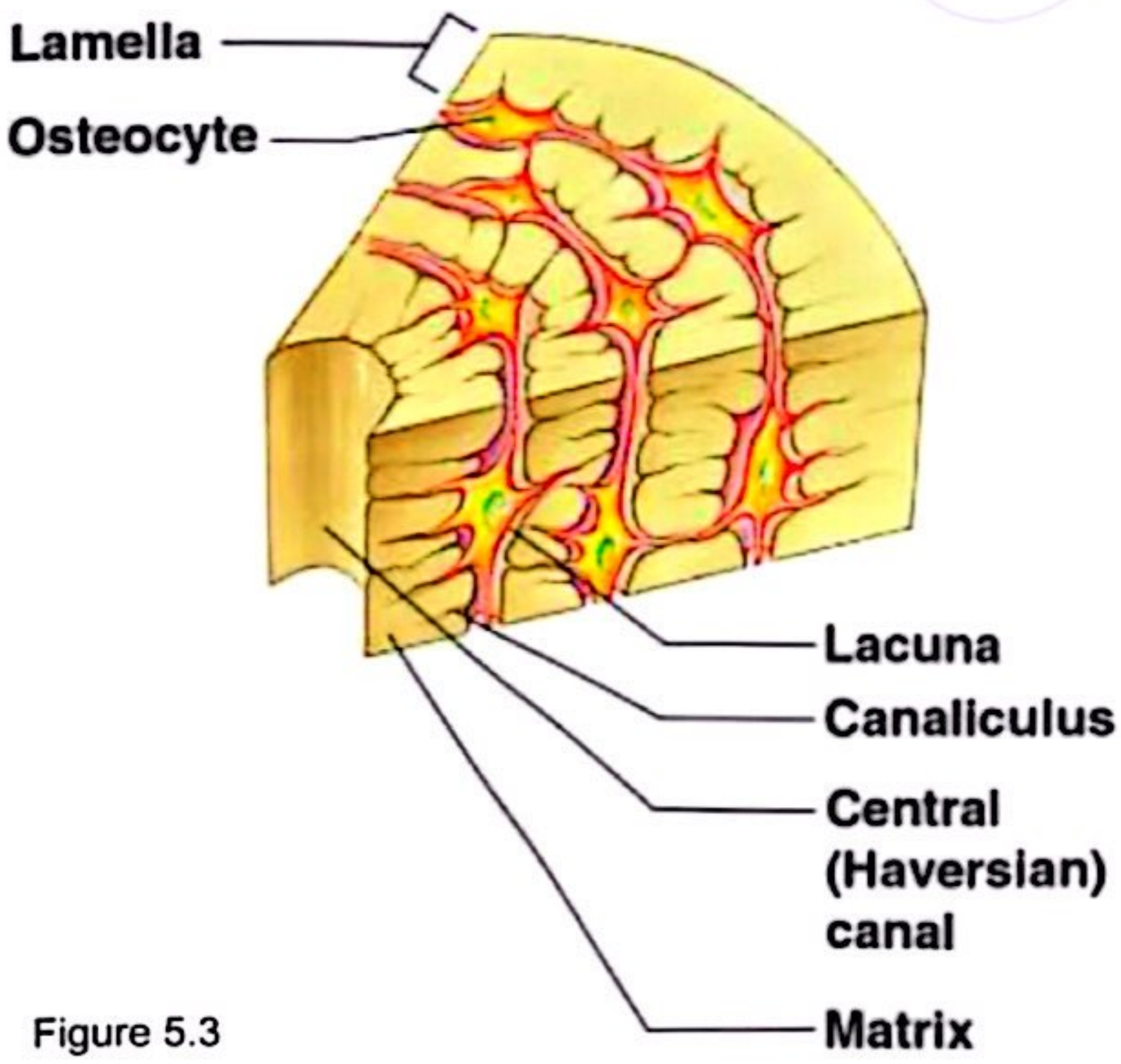


Figure 5.3

Microscopic Anatomy of Bone

- Canaliculi
  - Tiny canals
  - Radiate from the central canal to lacunae
  - Form a transport system

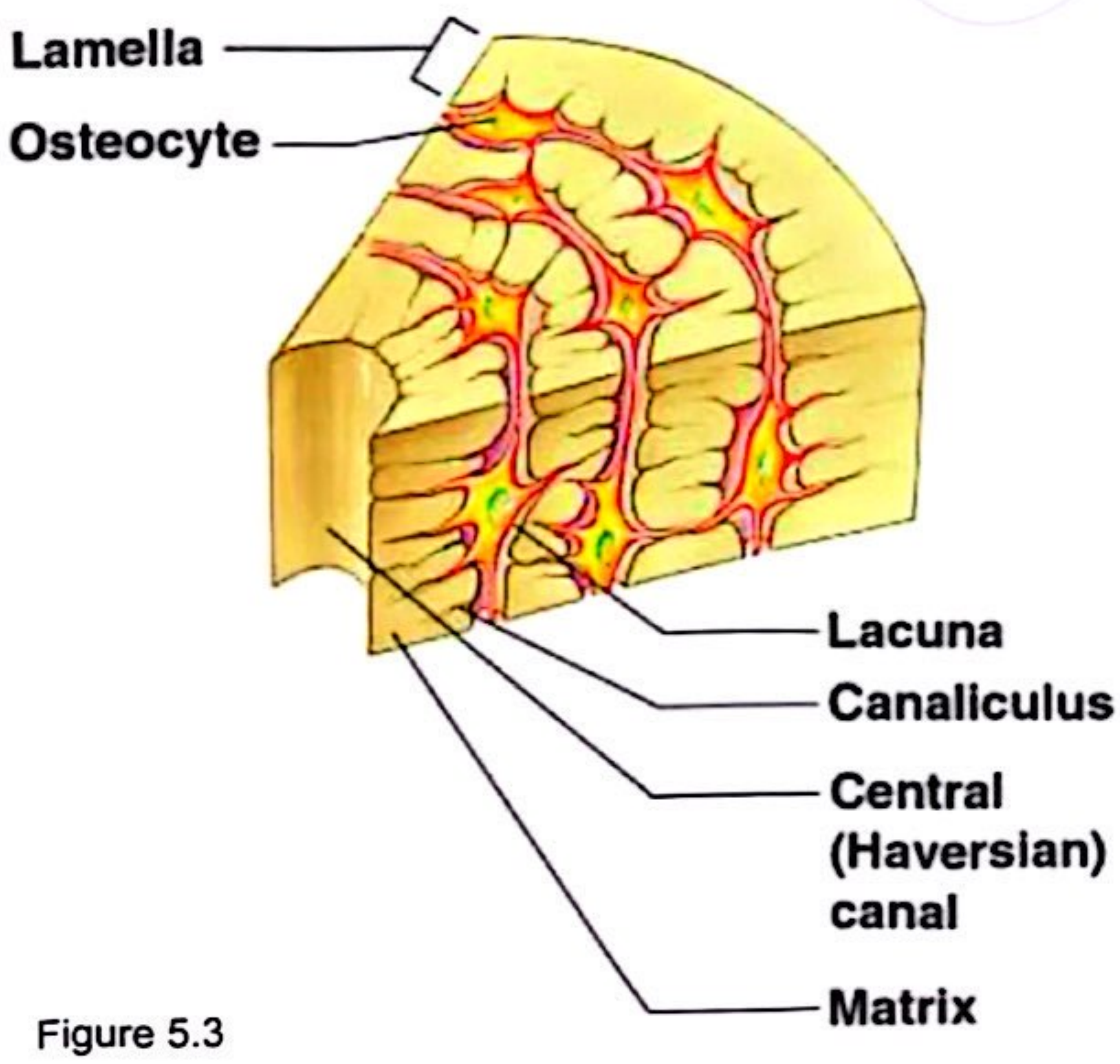


Figure 5.3

Changes in the Human Skeleton

- In embryos, the skeleton is primarily hya cartilage
- During development, much of this cartilage is replaced by bone
- Cartilage remains in isolated areas
  - Bridge of the nose
  - Parts of ribs
  - Joints

Bone Growth

- Epiphyseal plates allow for growth of long bone during childhood
  - New cartilage is continuously formed
  - Older cartilage becomes ossified
    - Cartilage is broken down
    - Bone replaces cartilage

Bone Growth

- Bones are remodeled and lengthened until growth stops
  - Bones change shape by gravity &muscle pull
  - Bones grow in width through periostium

Slide 5.13b

Long Bone Formation and Growt

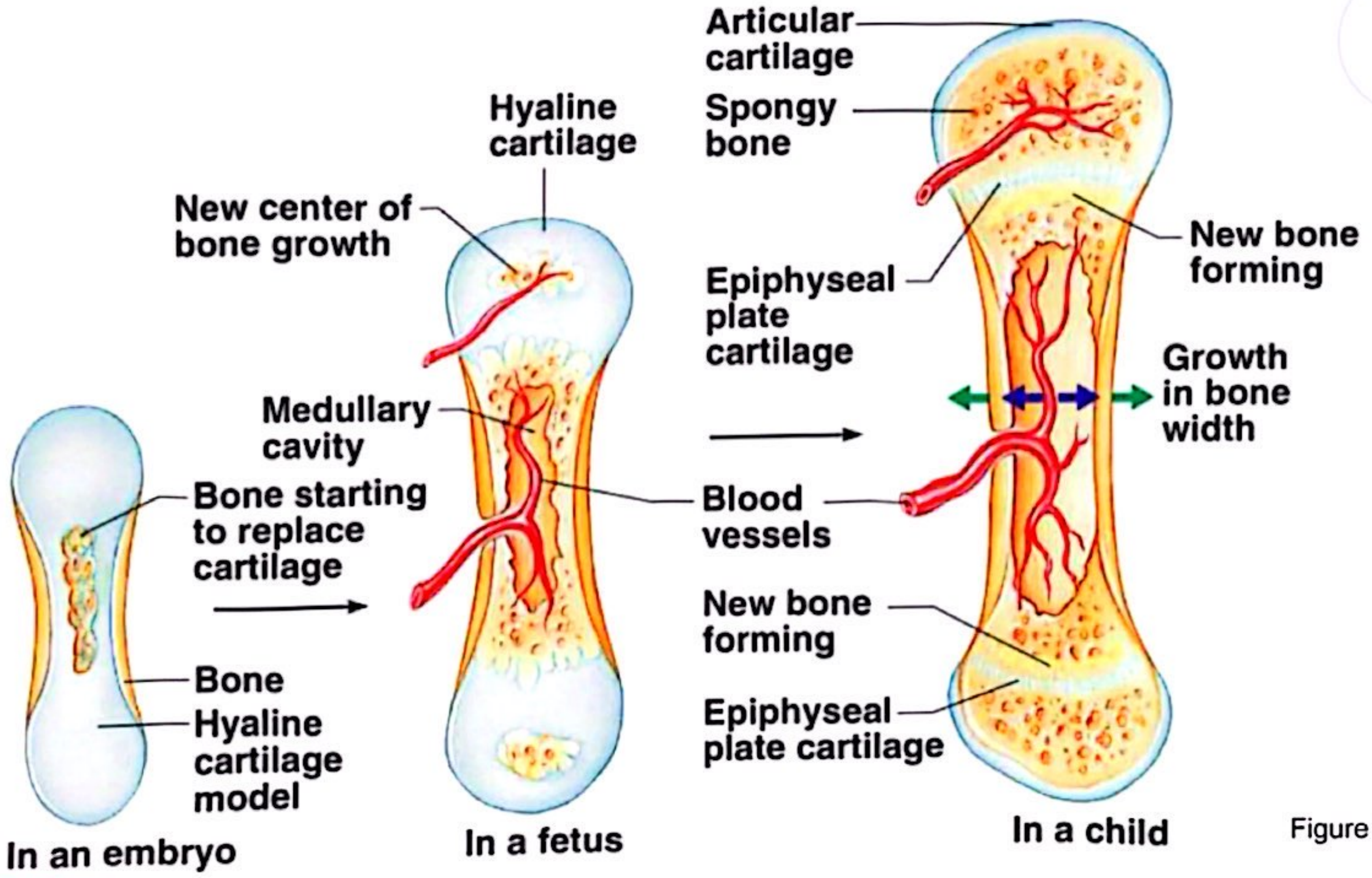


Figure 5.4a

Slide 5.14a

Types of Bone Cells

- Osteocytes
  - Mature bone cells
- Osteoblasts
  - Bone-forming cells
- Osteoclasts
  - Bone-destroying cells
  - Break down bone matrix for remodeling and release of calcium
- Bone remodeling is a process by both osteoblasts and osteoclasts

Fractures

- Closed fracture (simple): skin is intact
- Open fracture (compound): skin is open
- Fracture reduction :

  1-closed reduction ,no surgery is needed

  2-open reduction ,surgery is needed

Repair of fracture

- Healing time for simple fracture is 6-8 wee (longer in elderly people)
- It occurs in FOUR major events
- 1-hematoma formation
- 2-fibrocartilage callus formation
- 3-bony callus formation
- 4-bone remodelling

Skeletal system includes

- Axial division
  - Skull and associated bones
    - Auditory ossicles
    - Hyoid bones
  - Vertebral column
  - Thoracic cage(Ribs+ sternum)

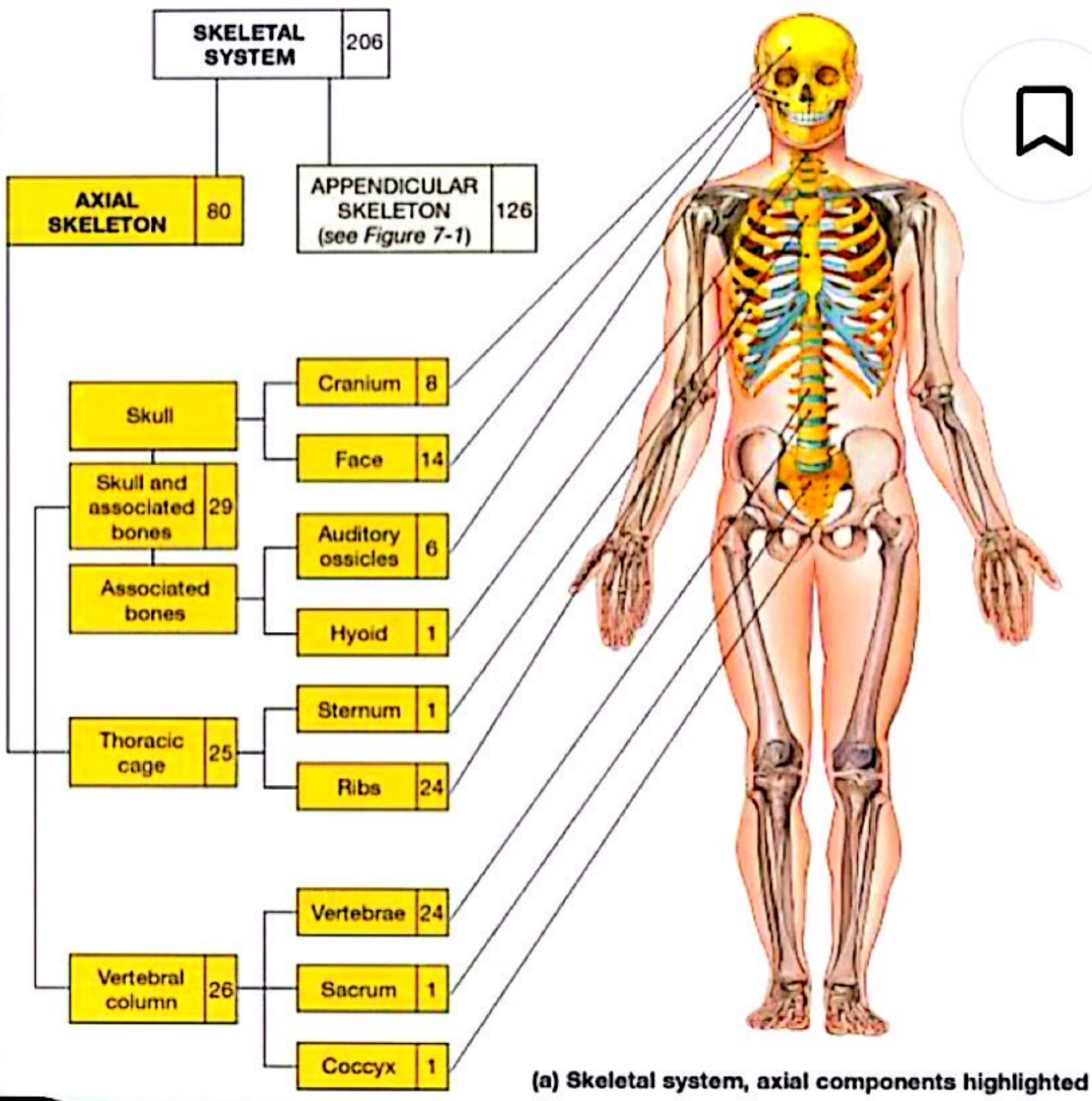
- Appendicular division

  -Pectoral girdle

  -Pelvic girdle

The Axial Skeleton

- Axial division
  - Skull and associa
    bones:
    - Auditory ossicles
    - Hyoid bones
  - Vertebral columr
  - Thoracic cage
    - Ribs + sternum



(a) Skeletal system, axial components highlighted

The Skull and Associated Bones

Cranial Bones		Facial Bones	
Occipital bone	Single	Vomer	Single
Frontal bone	Single	Lacrimal bones	Paired
Parietal bones	Paired	Nasal bones	Paired
Temporal bones	Paired	Inferior nasal conchae	Paired
Sphenoid bone	Single	Zygomatic bones	Paired
Ethmoid bone	Single	Maxillary bones	Paired
		Mandible	Single

*Note that there are single and paired bones.

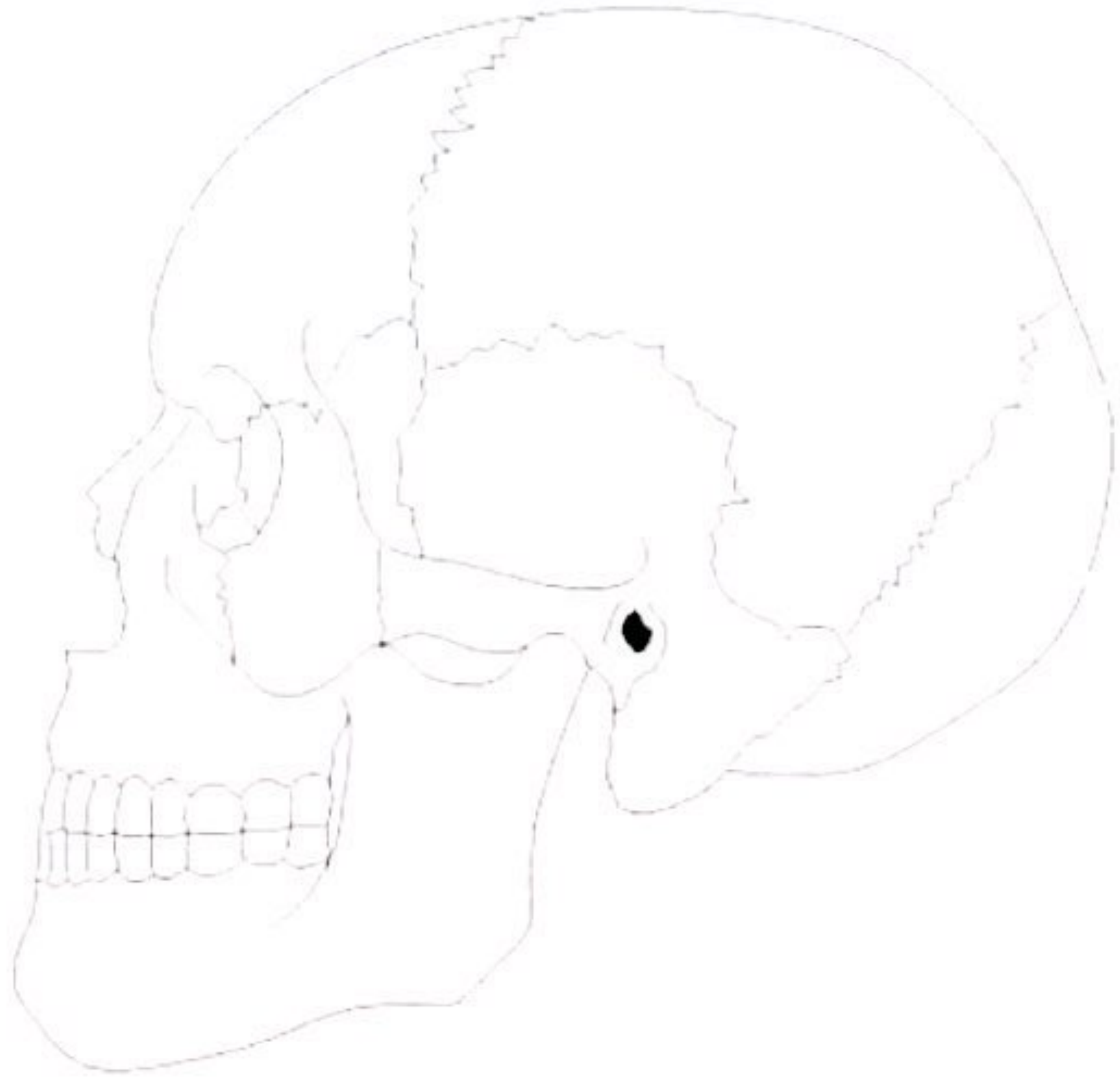
Cranial and facial bones. (Table 3-1)

Sutures

- Sutures – Immovable joints that join skull bones together
- Form boundaries between skull bones
- Four sutures:
  - Coronal – between parietal and frontal
  - Sagittal– between parietal bones
  - Lambdoid – between the parietal and occipital
  - Squamous – between the parietal and temporal
  - ✓Fontanels – usually ossify by 2 years of age

Sagittal

Frontal
(Coronal)



Squamous

Lambdoid

Sutures

The Adult Skull

- skull = 22 bones
- ***cranium*** = 8 bones: frontal, occipital, 2 temporals, 2 parietals, sphenoi ethmoid
- ***facial bones*** = 14 bones: nasals, maxillae, zygomatics, mandible, lacrimals, palatines, inferior nasal conchae, vomer.

- skull forms a larger ***cranial cavity***
  - -also forms the ***nasal cavity, the orbits, paranasal sinuses***
  - mandible and ***auditory ossicles*** are the only movable skull bones

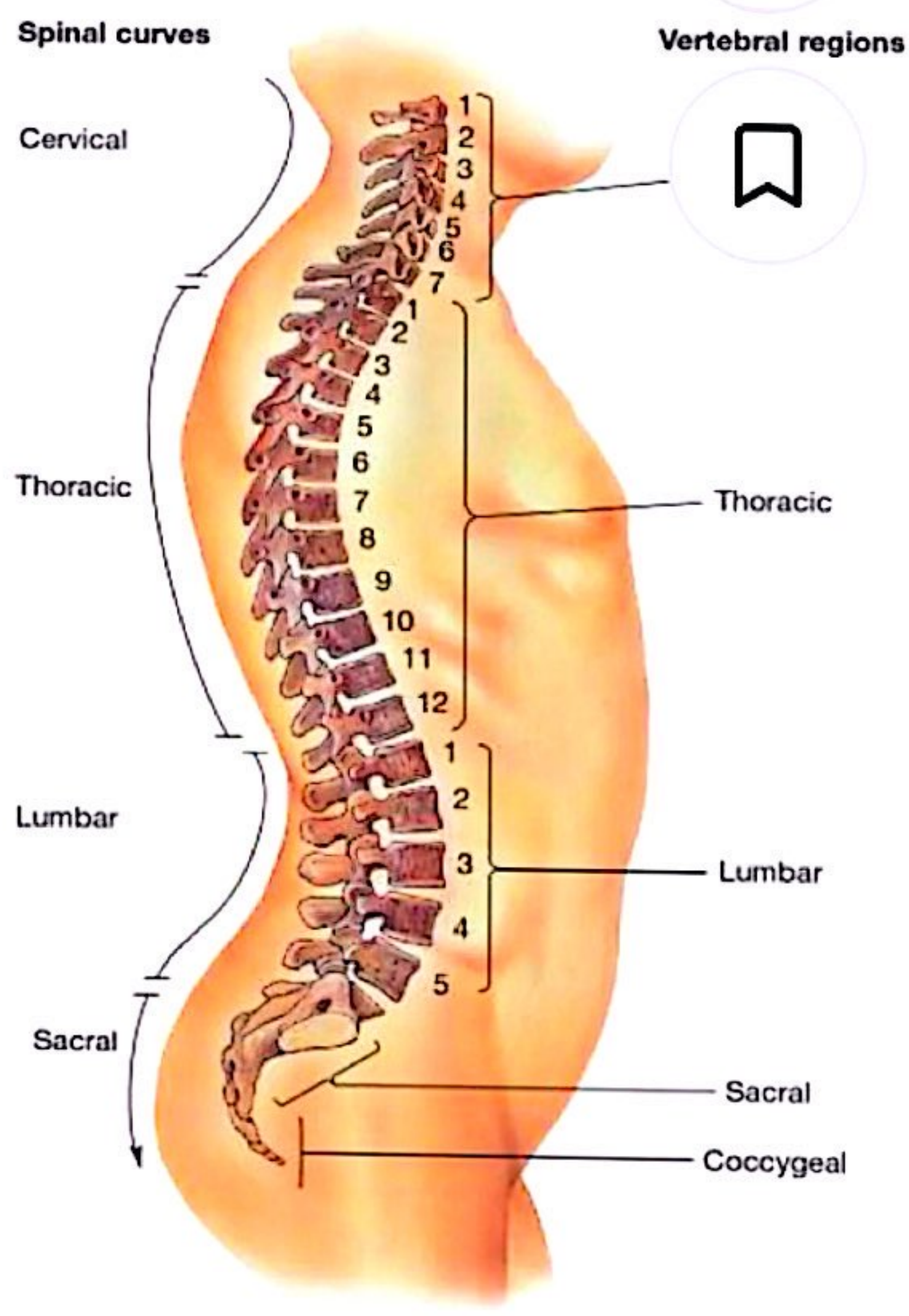
- cranial bones also: attach to membranes called ***meninges***
  - -stabilize positions of the brain, blood vessels
  - -outer surface provides large areas for muscle attachment that move the head or provide facial expressions

The Vertebral Column

http://www.wisc-online.com/objects/index.asp?objID=AP12104

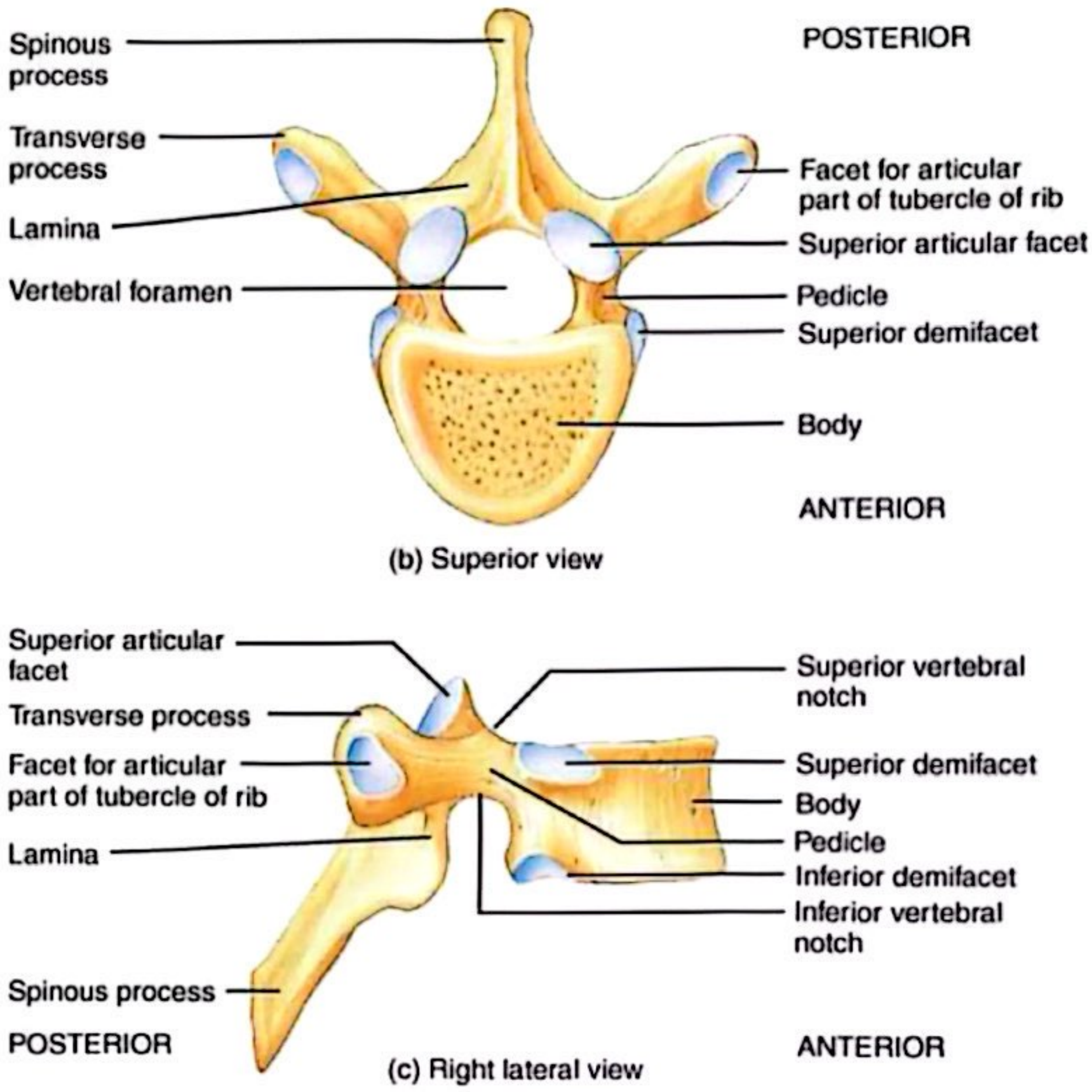
Adult Vertebral Column

- 26 vertebrae
  - 24 individual vertebrae
  - Sacrum
  - Coccyx
- Seven cervical vertebrae
- Twelve thoracic vertebrae
- Five lumbar vertebrae
- Sacrum and coccyx are
- Fused together.



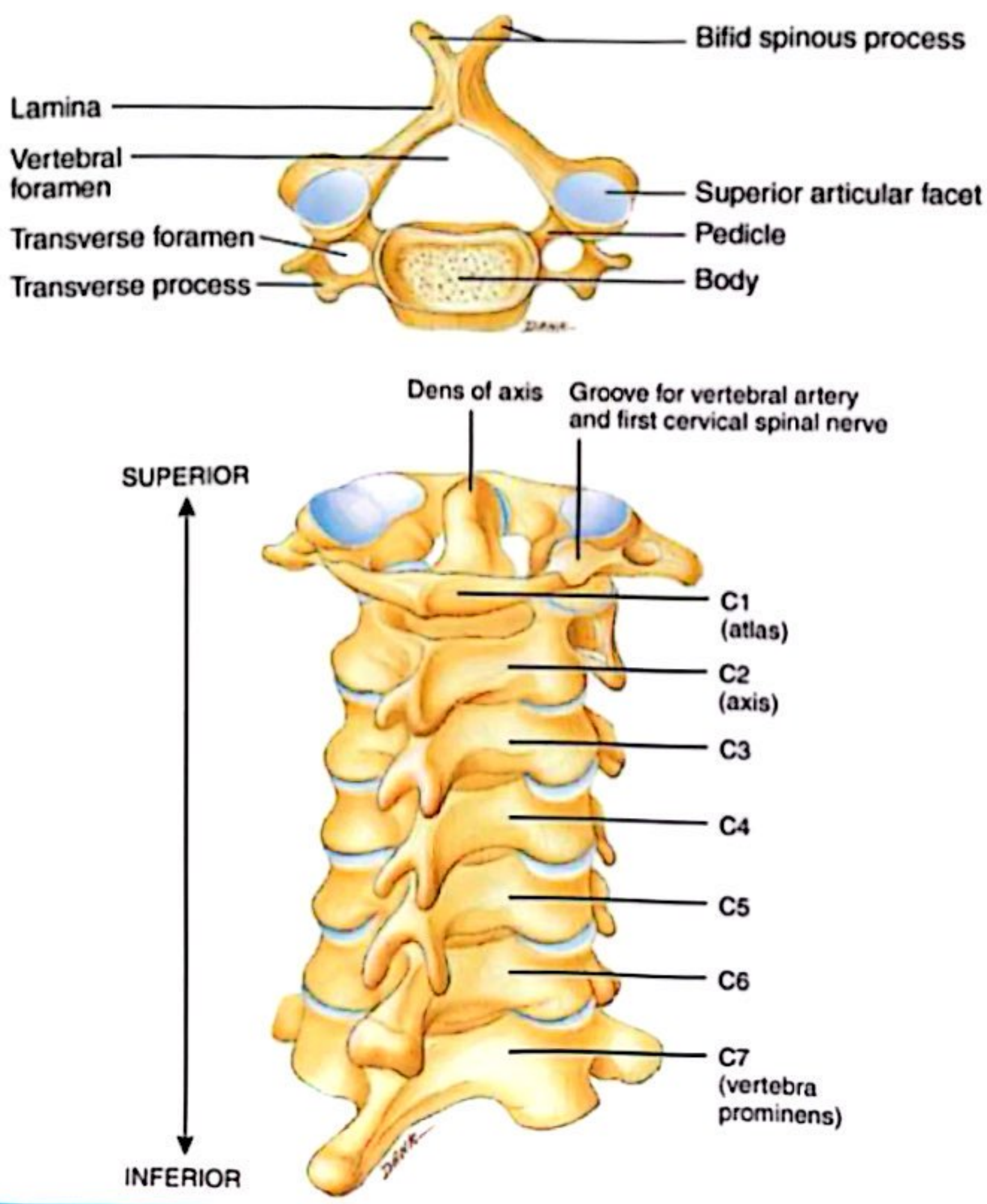
(a) Lateral view

Typical Vertebrae

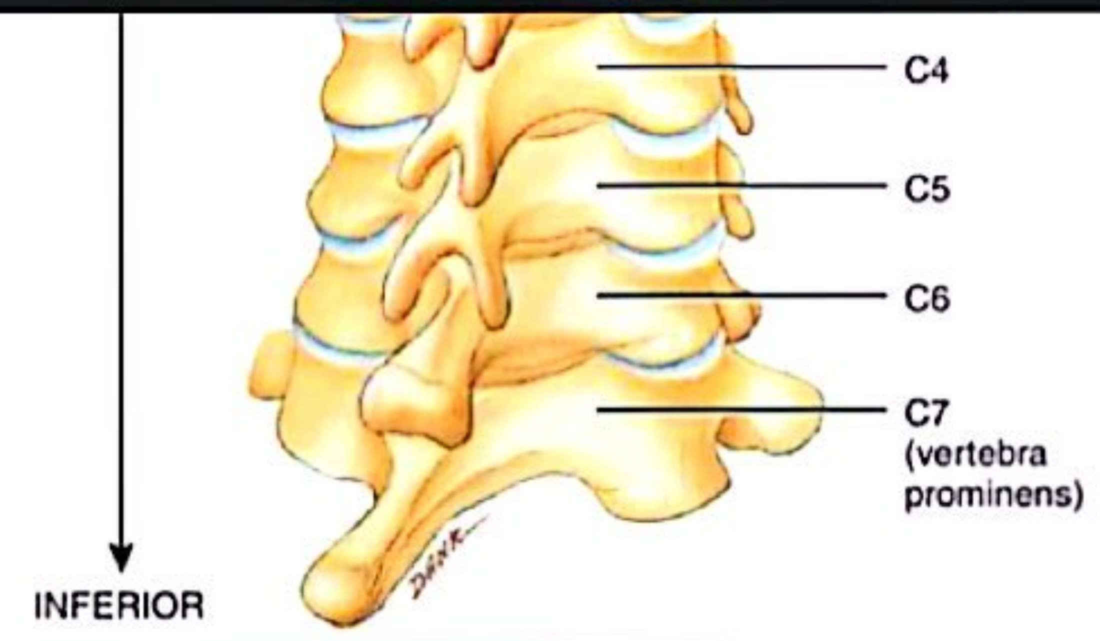


- Body
  - weight bearing
- Vertebral arc
  - pedicles
  - laminae
- Vertebral foramen
- Seven processes
  - 2 transverse
  - 1 spinous
  - 4 articular

Typical Cervical Vert. (C3–C7)

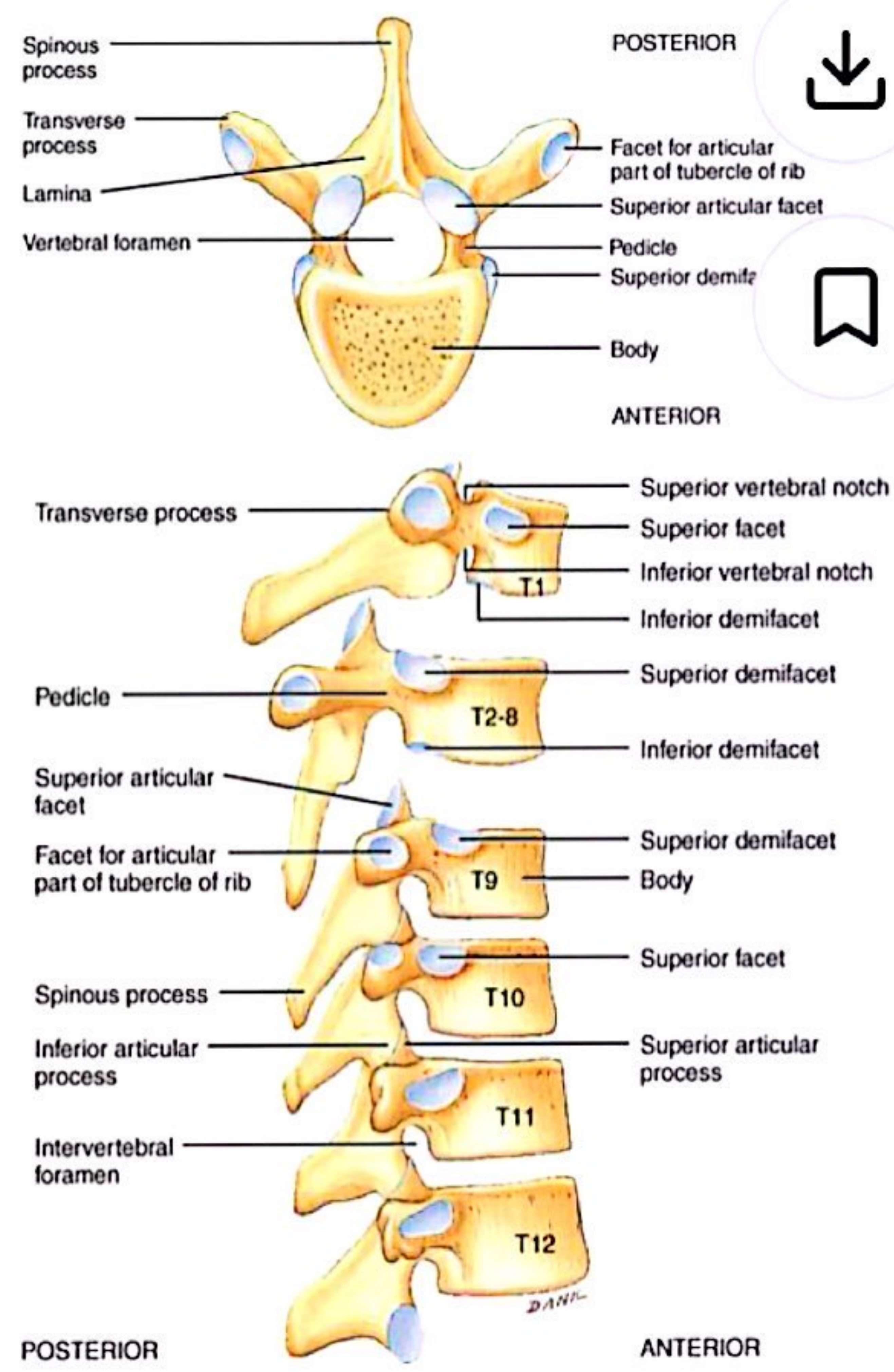


- Smaller bodies
- Larger spinal canal
- 1st and 2nd cervical vertebrae are unique
  - atlas & axis



horacic Vertebrae T1–T12)

- All articulate with ribs
- Have heart-shaped bodies
- Each side of the body bears demifacets for articulation with ribs

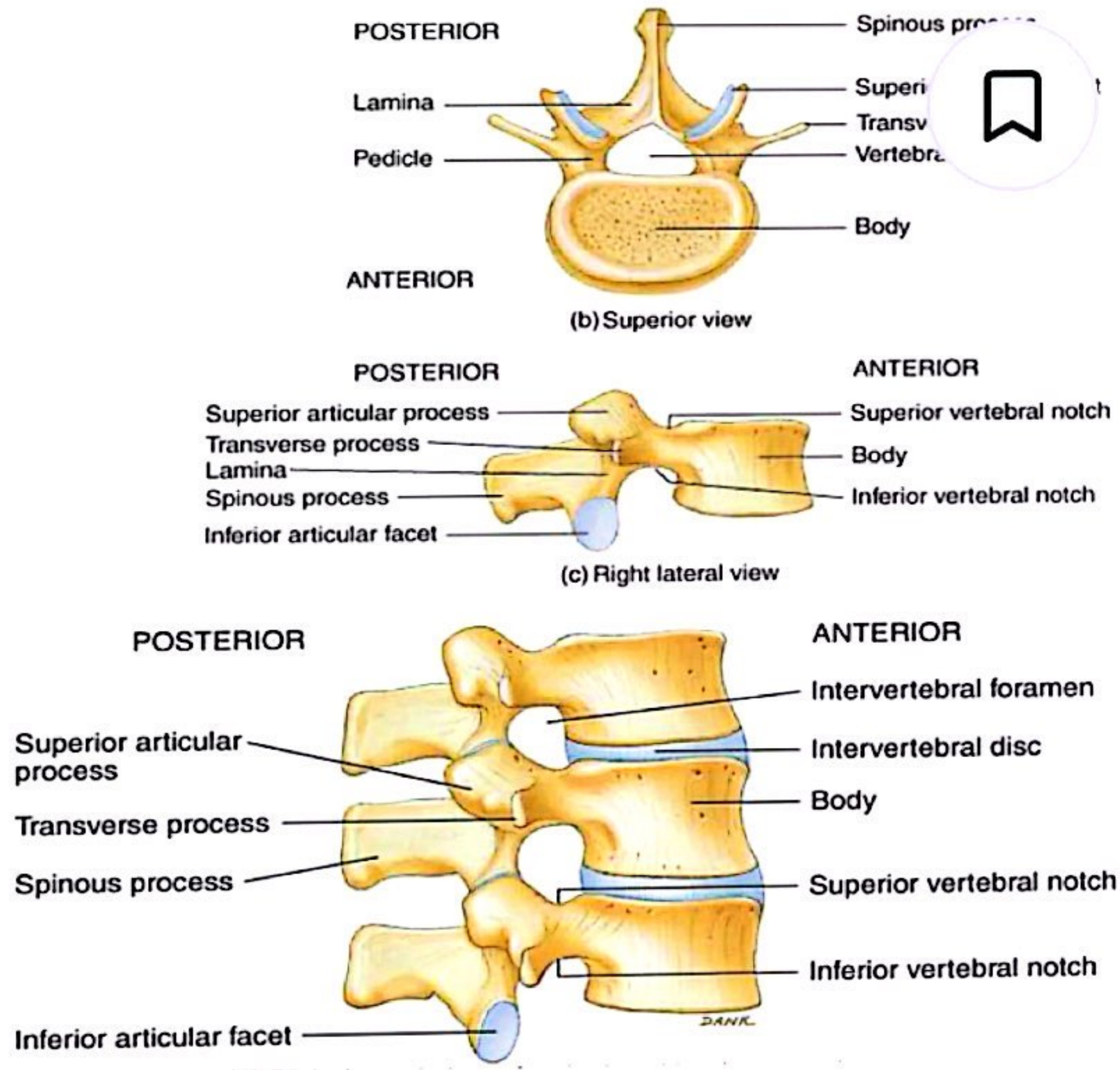


Thoracic Vertebrae

- Allows rotation and prevents flexion and extension

Lumbar Vertebrae

- Bodies are thick and strong
- Allows flexion and extension – rotation prevented



Sacrum (S_1 – S_5)

- Forms the posterior wall of pelvis
- Formed from 5 fused vertebrae
- Superior surface articulates with L_5
- Inferiorly articulates with coccyx

Sacrum

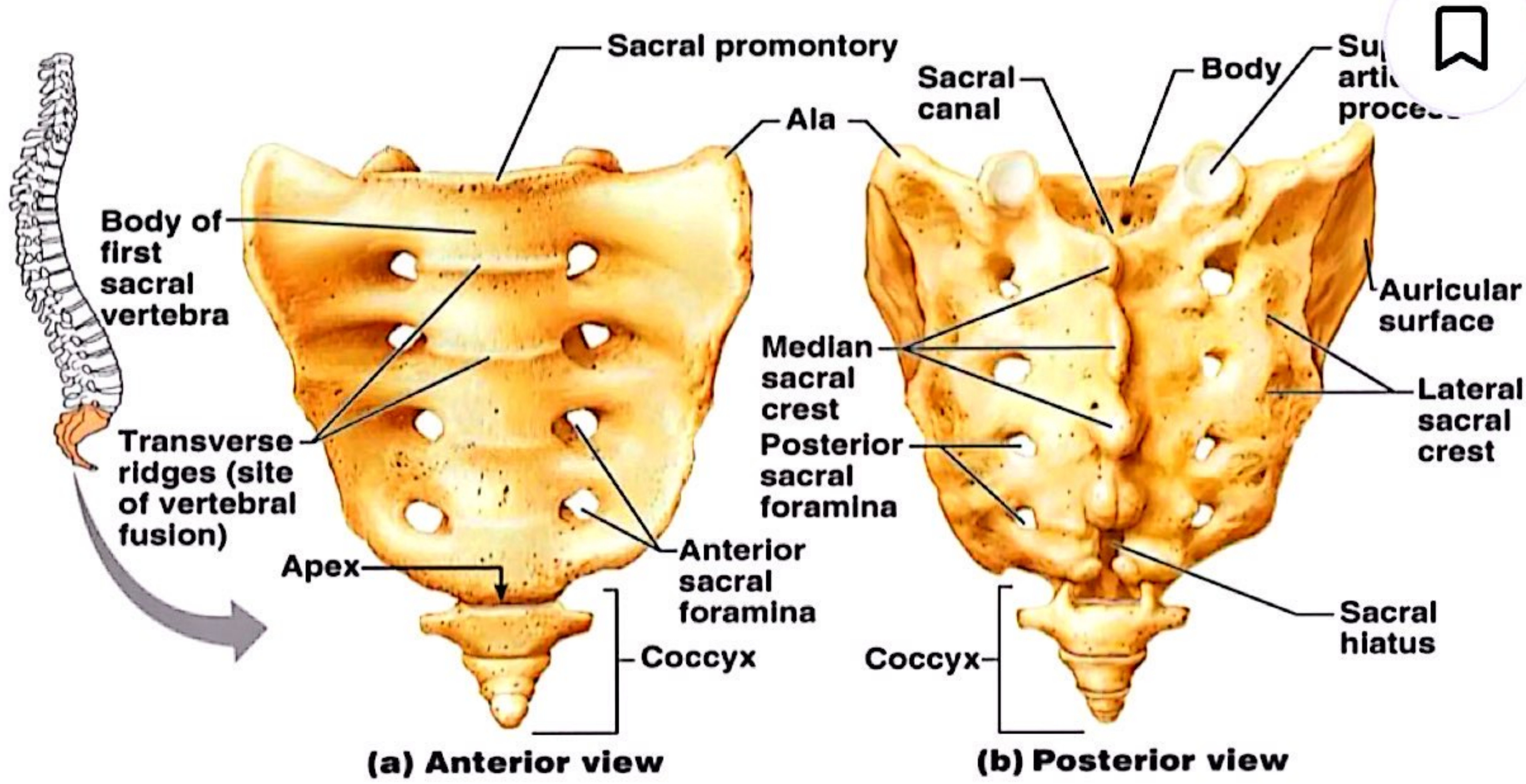


Figure 7.18a, b

Coccyx

- Is the "tailbone"
- Formed from 3 – 5 fused vertebrae
- Offers only slight support to pelvic organs

Bony Thorax

- Forms the framework of the chest
- Components of the bony thorax
  - Thoracic vertebrae – posteriorly
  - Ribs – laterally
  - Sternum and costal cartilage – anteriorly
- Protects thoracic organs
- Supports shoulder girdle and upper limbs
- Provides attachment sites for muscles

The Bony Thorax

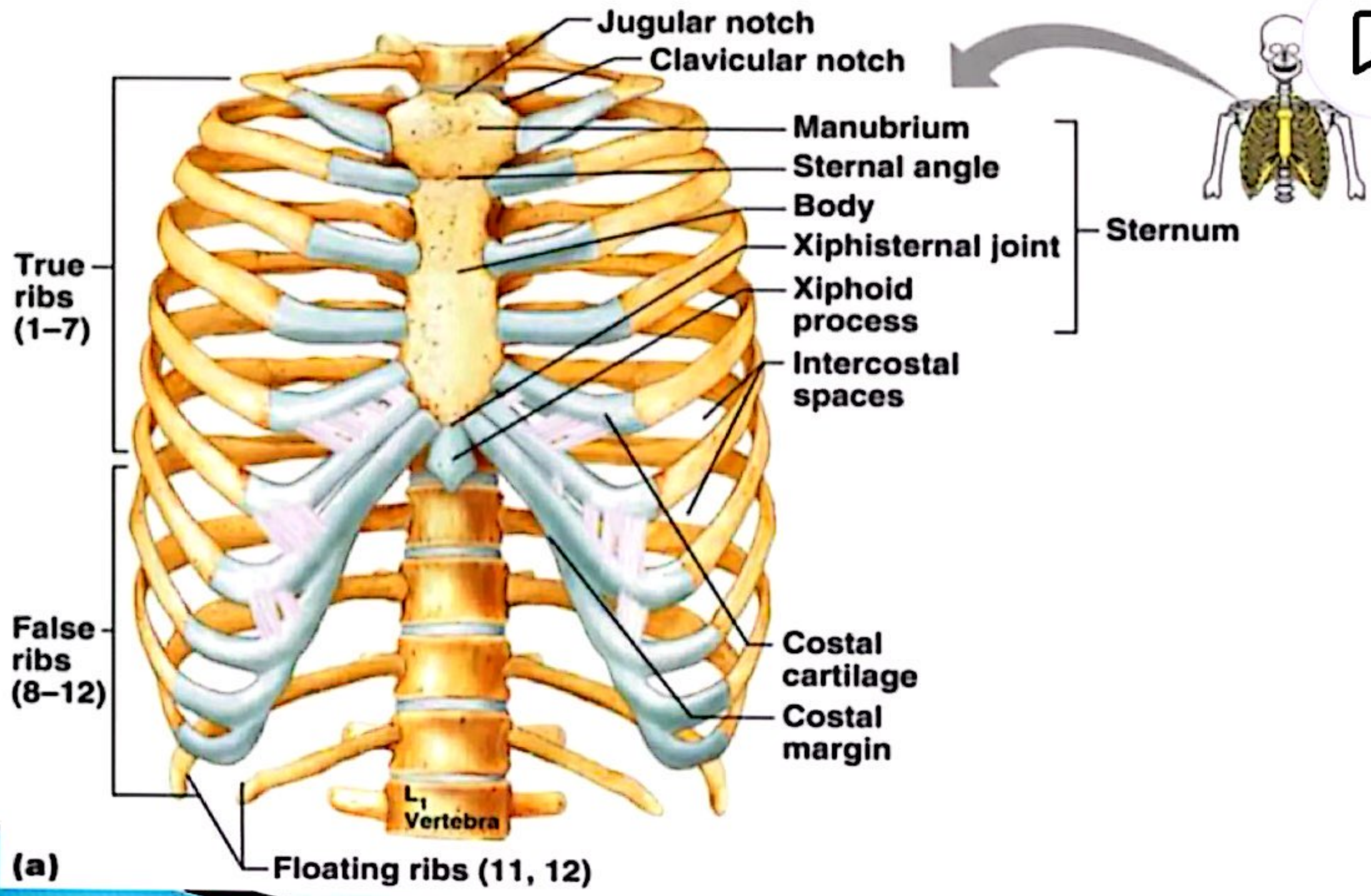


Figure 7.19a

Sternum

- Formed from three parts :
  - **Manubrium** – superior part
    - Articulates with medial end of clavicles
  - **Body** – bulk of sternum
    - Sides are articulate for costal cartilage of ribs 2-7
  - **Xiphoid process** – inferior end of sternum
    - Ossifies around age 40

Ribs

- All ribs attach to vertebral column posterio
  - **True ribs** – superior seven pairs of ribs
    - Attach to sternum by costal cartilage
  - **False ribs** – inferior five pairs of ribs ,attach indirectly to the sternum
  - **floating ribs** ribs 11-12 are short and free anteriorly.

Disorders of the Axial Skeleto

- Abnormal spinal curvatures
  - **Scoliosis** – an abnormal lateral curvature
  - **Kyphosis** – an exaggerated thoracic curvature
  - **Lordosis** – an inward lumbar curvature – “swayback”
- Stenosis of the lumbar spine
  - A narrowing of the vertebral canal

The Appendicular Skeleton

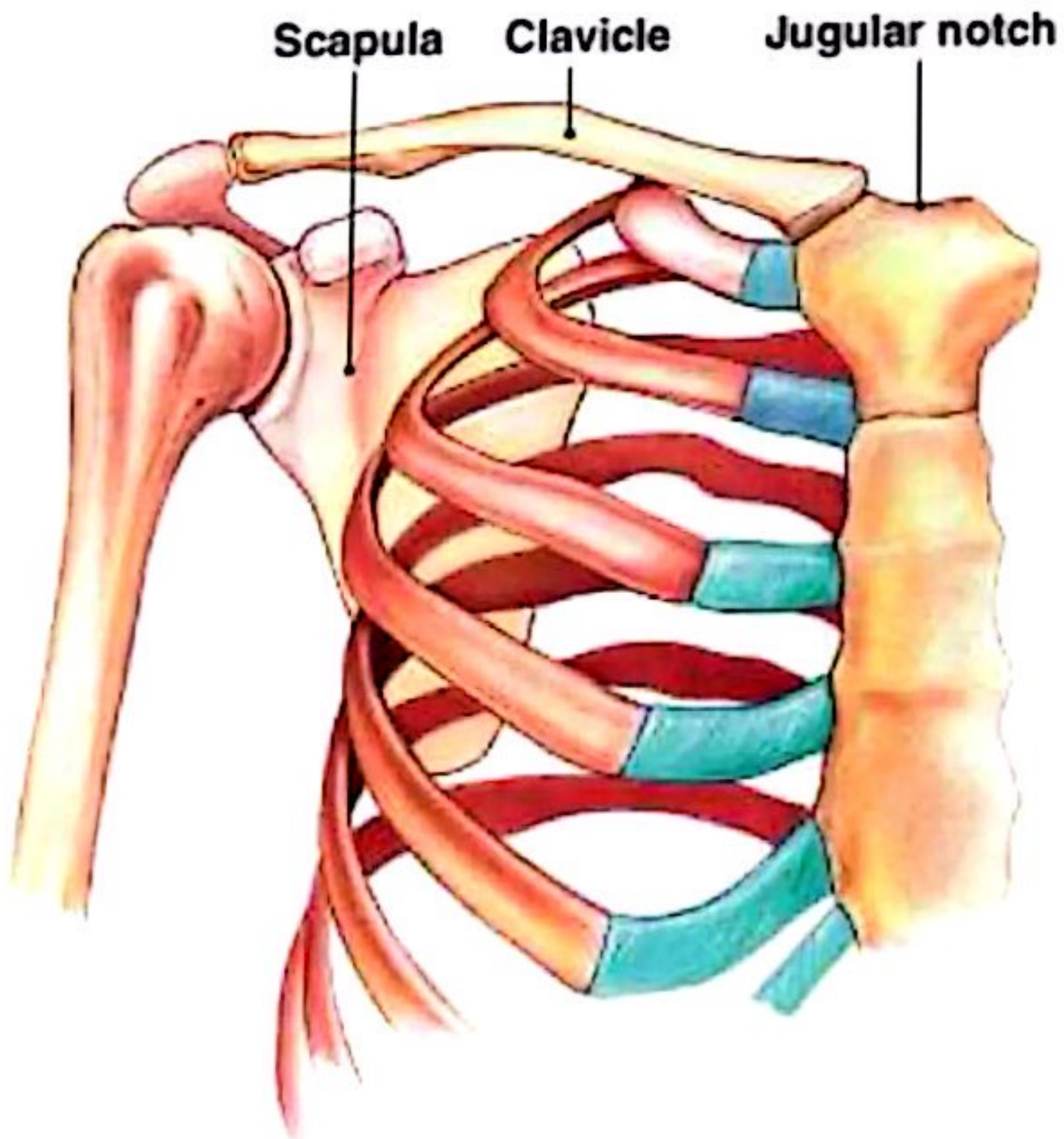
- Allows us to move and manipulate objects
- Includes all bones other than axial skeleton, it includes:
  - the limbs (upper & lower limbs)
  - the supportive girdles (pectoral &pelvic girdles)

What are the bones of the pectoral girdle, their functions, and features?

The Pectoral Girdle

- Also called the shoulder *girdle*
- Connects the arms to the body
- Positions the shoulders
- Provides a base for arm movement

The Pectoral Girdle



(a) Anterior view

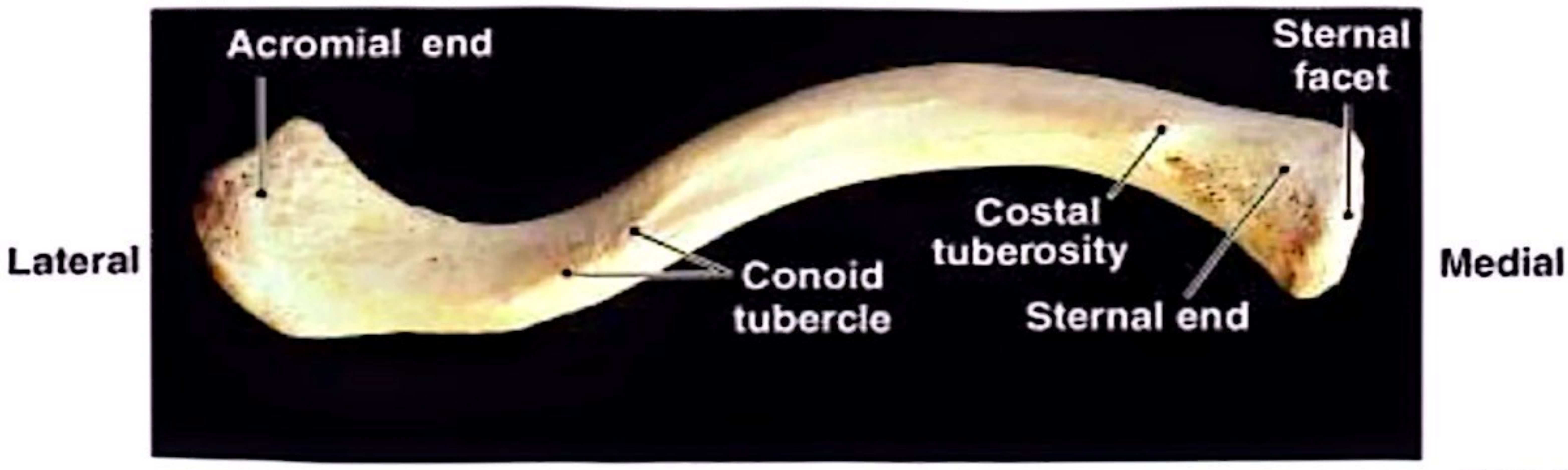
Figure 8-2a

The Pectoral Girdle

- Consists of:
  - 2 clavicles
  - 2 scapulae
- Connects with the axial skeleton only at the manubrium(claviculosternal joint)

The Clavicles

(b) Superior view



(c) Inferior view

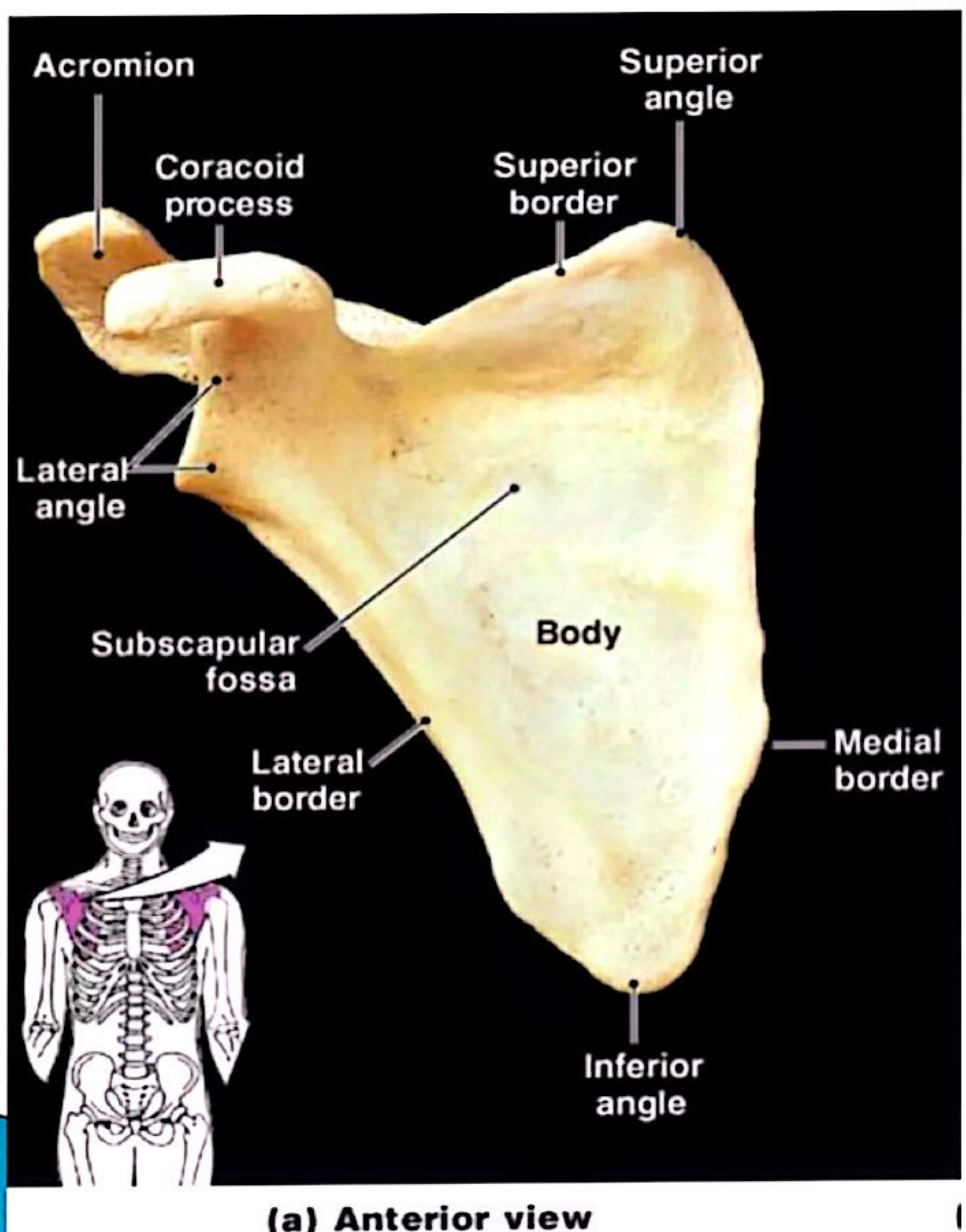
Figure 8-2b, c

The Clavicles

- Also called *collarbones*
- Long, S-shaped bones
- Originate at the manubrium (sternal end)
- Articulate with the scapulae (acromial end)

The Scapulae

- Also called *shoulder blades*
- Broad, flat and triangular
- Articulate with arms and collarbone



Anatomy of The scapula

Figure 8-3a

What are the bones of the upper limbs, their functions, and features?

The Upper Limbs

- Arms, forearms, wrists, and hands

  Note: arm (*brachium*) = 1 bone, the humerus

e Humerus

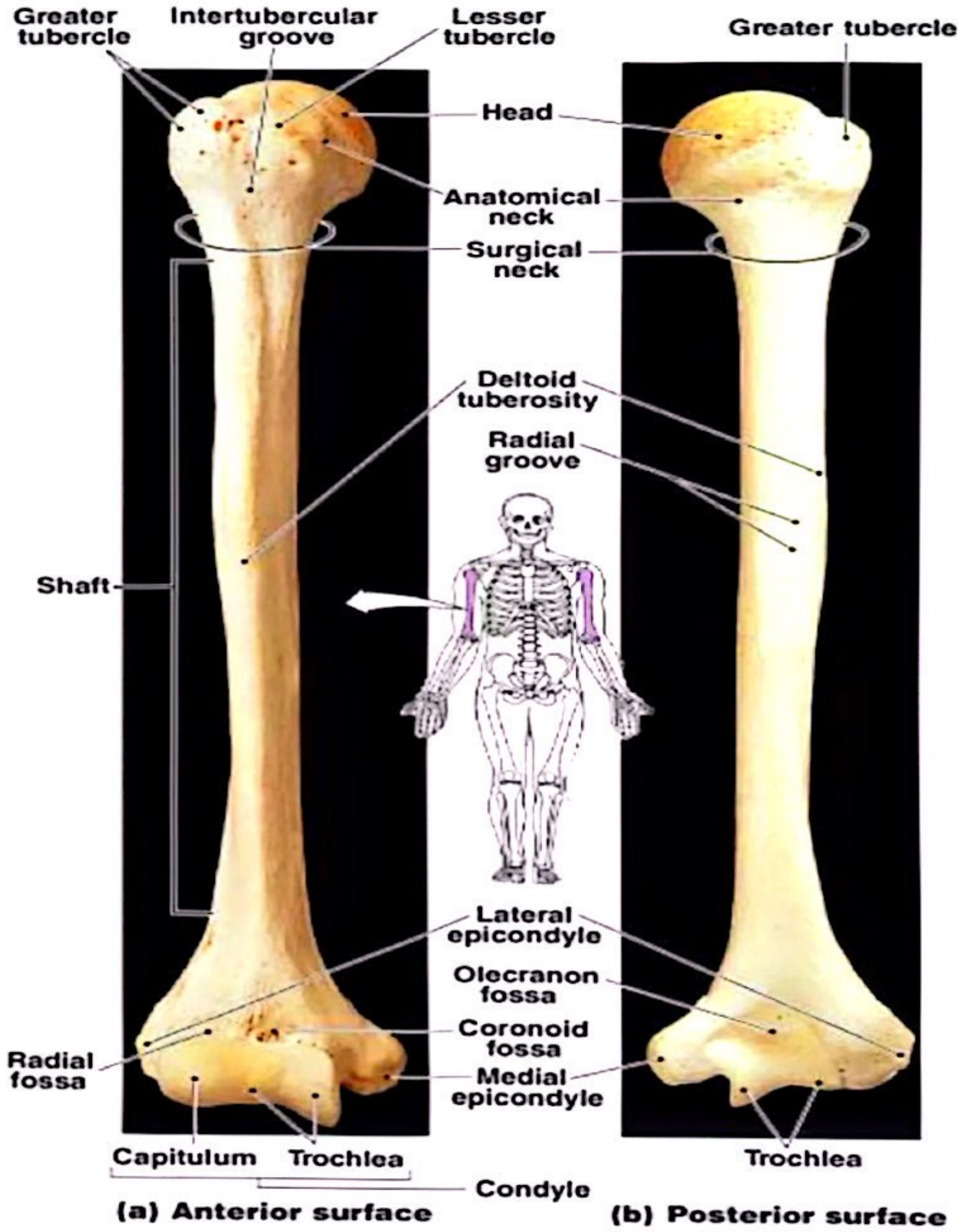


Figure 8-4

The Humerus

- Also called the *arm*
- The long, upper armbone
- Articulates with the pectoral girdle

The Forearm

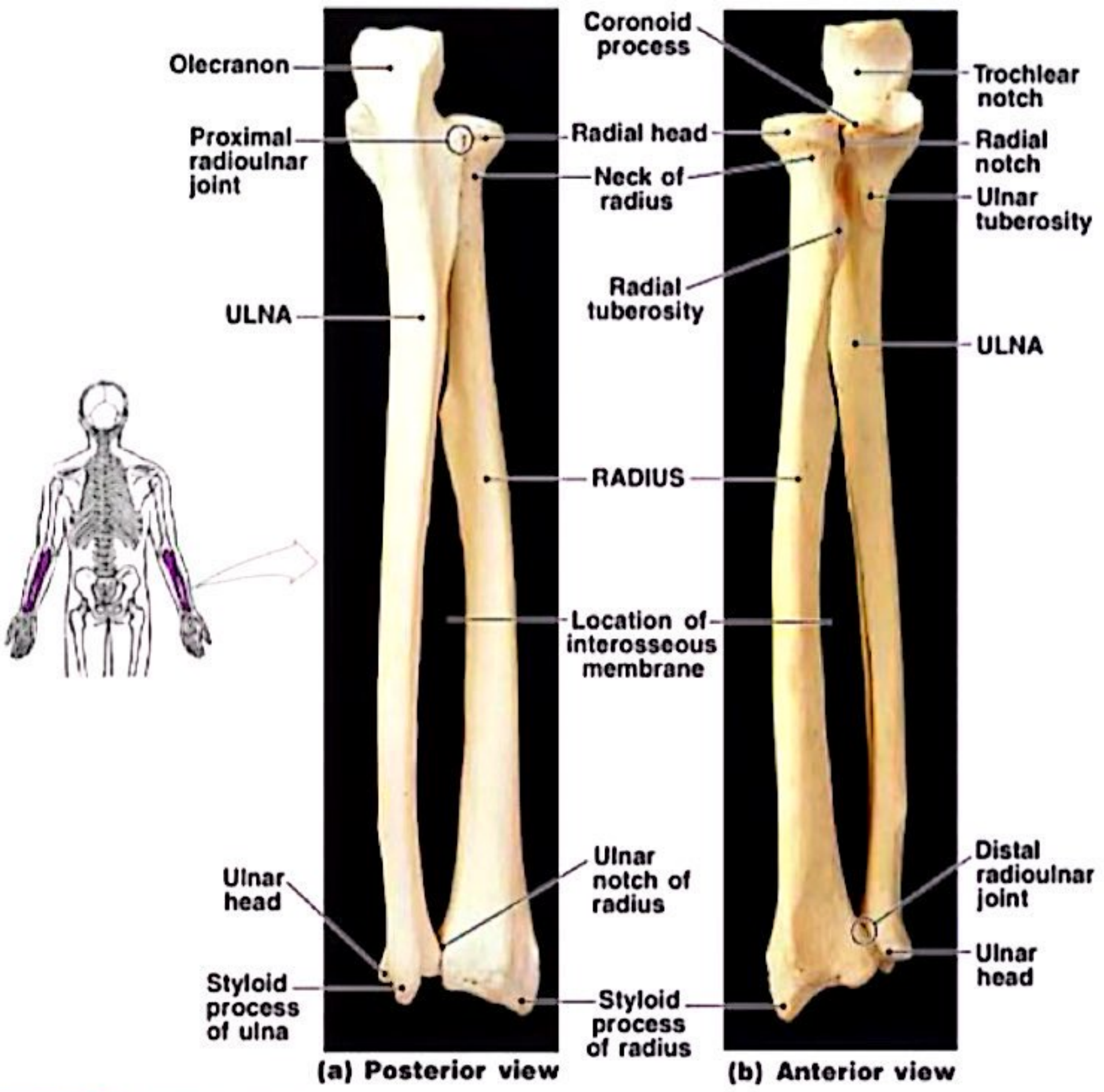


Figure 8-5

The Forearm

- Also called the *antebrachium*
- Consists of 2 long bones:
  - ulna (medial)
  - radius (lateral)

The Wrist

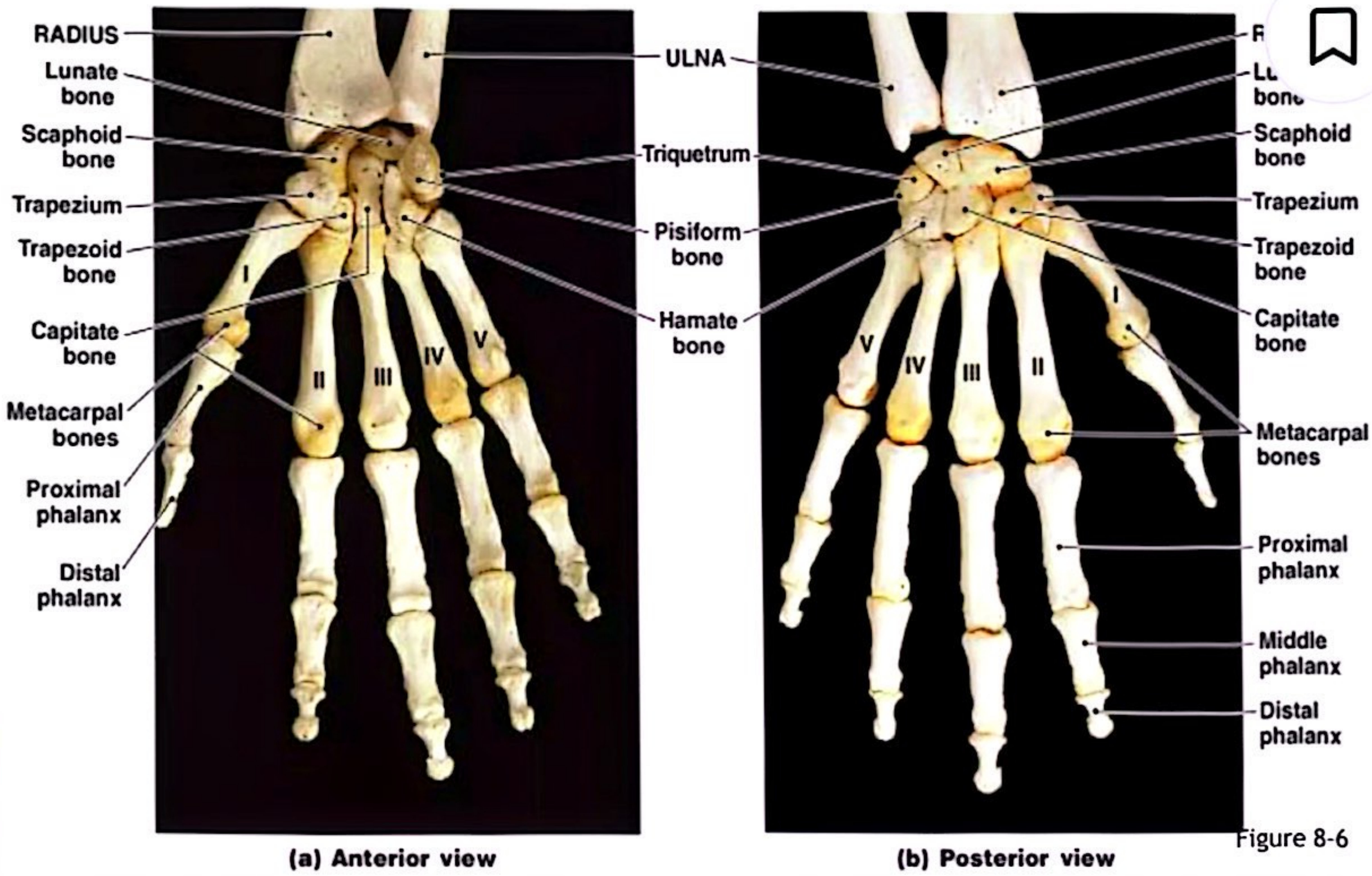


Figure 8-6

The Wrist

- 8 carpal bones:
  - 4 proximal carpal bones
  - 4 distal carpal bones
  - allow wrist to bend and twist

Metacarpal Bones

- The 5 long bones of the hand
- Numbered I–V from lateral (thumb) to medial
- Articulate with proximal phalanges

Phalanges of the Hands

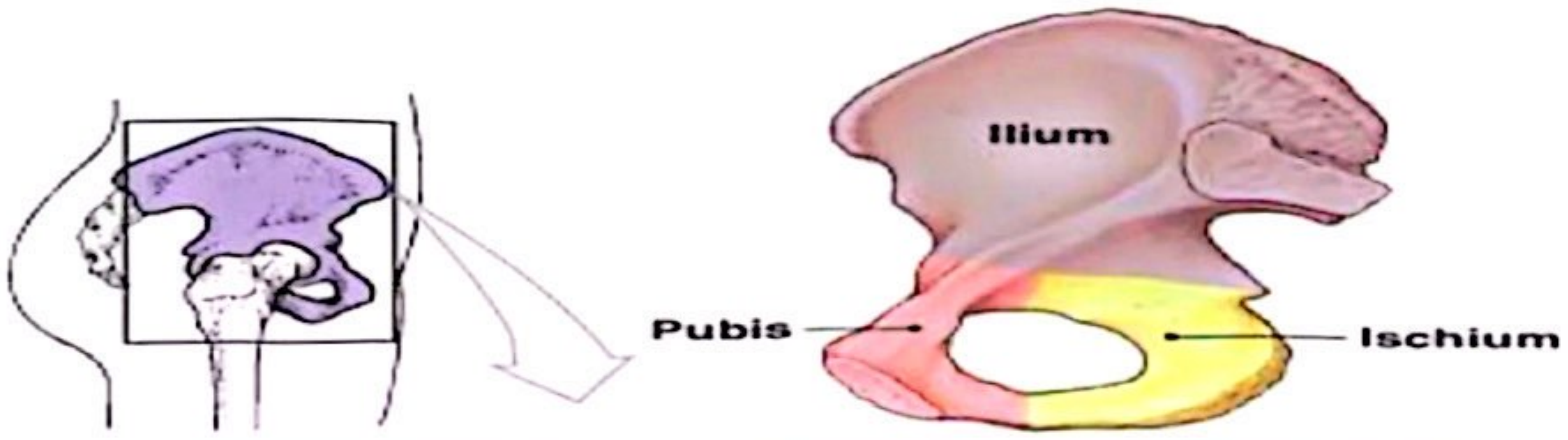
- Thumb:
  - 2 phalanges (proximal, distal)
- Fingers:
  - 3 phalanges (proximal, middle, distal)

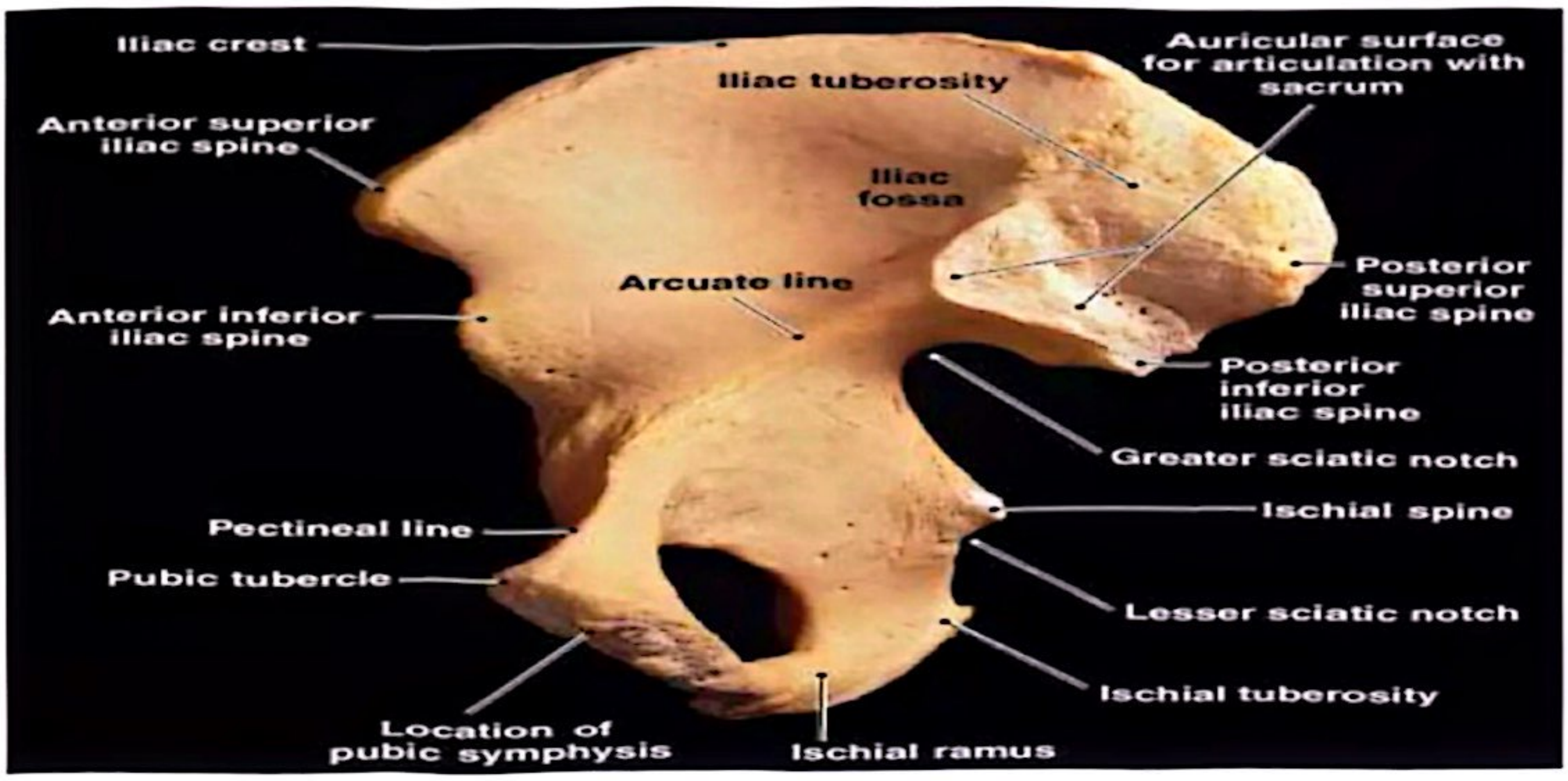
What are the bones of the pelvic girdle, their functions, and features?

The Pelvis

- Consists of 2 ossa coxae, the sacrum, and the coccyx
- Stabilized by ligaments of pelvic girdle, sacrum, and lumbar vertebrae

The Pelvic Girdle





(b) Right os coxae, medial view

-7

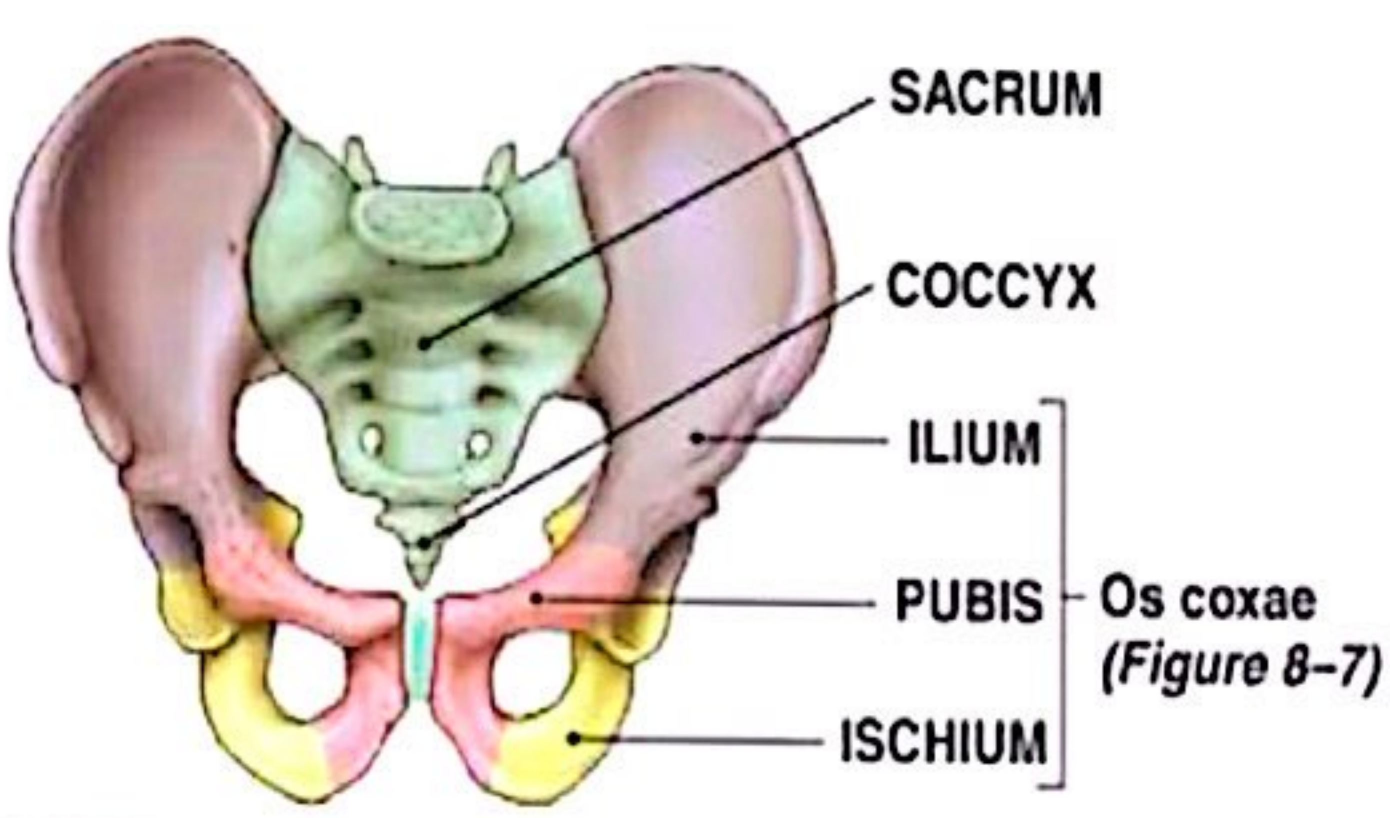
The OSSA COXAE

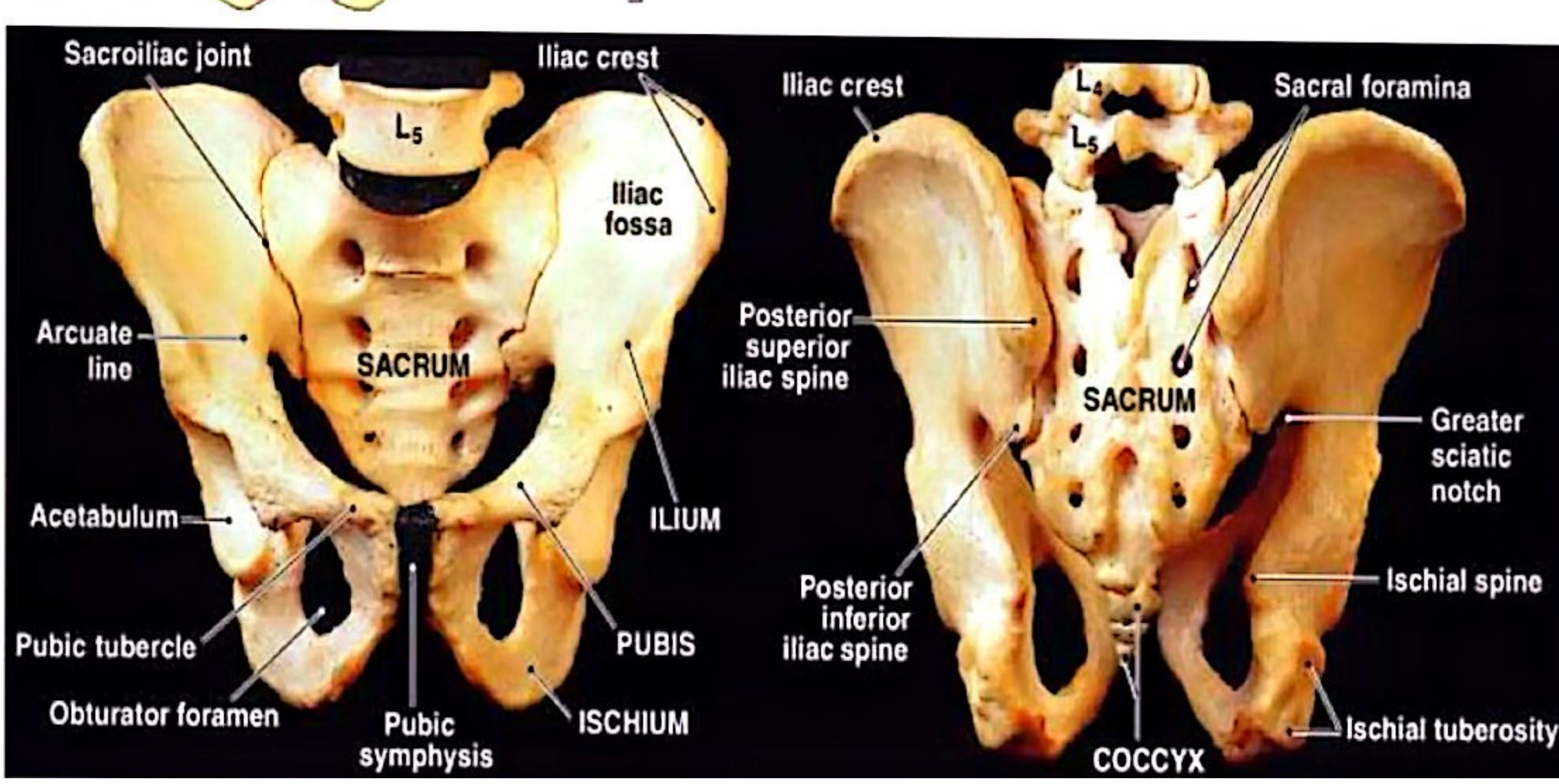
- Also called hipbones
- Strong to bear body weight &stress of movement
- Each is made up of 3 fused bones:
  - ilium (articulates with sacrum)
  - ischium
  - pubis

The Acetabulum (vinegar cup)

- Also called the *hip socket*
- Is the meeting point of the ilium, ischium, and pubis
- Articulates with head of the femur (Hip joint))

The Pelvis





Divisions of the Pelvis

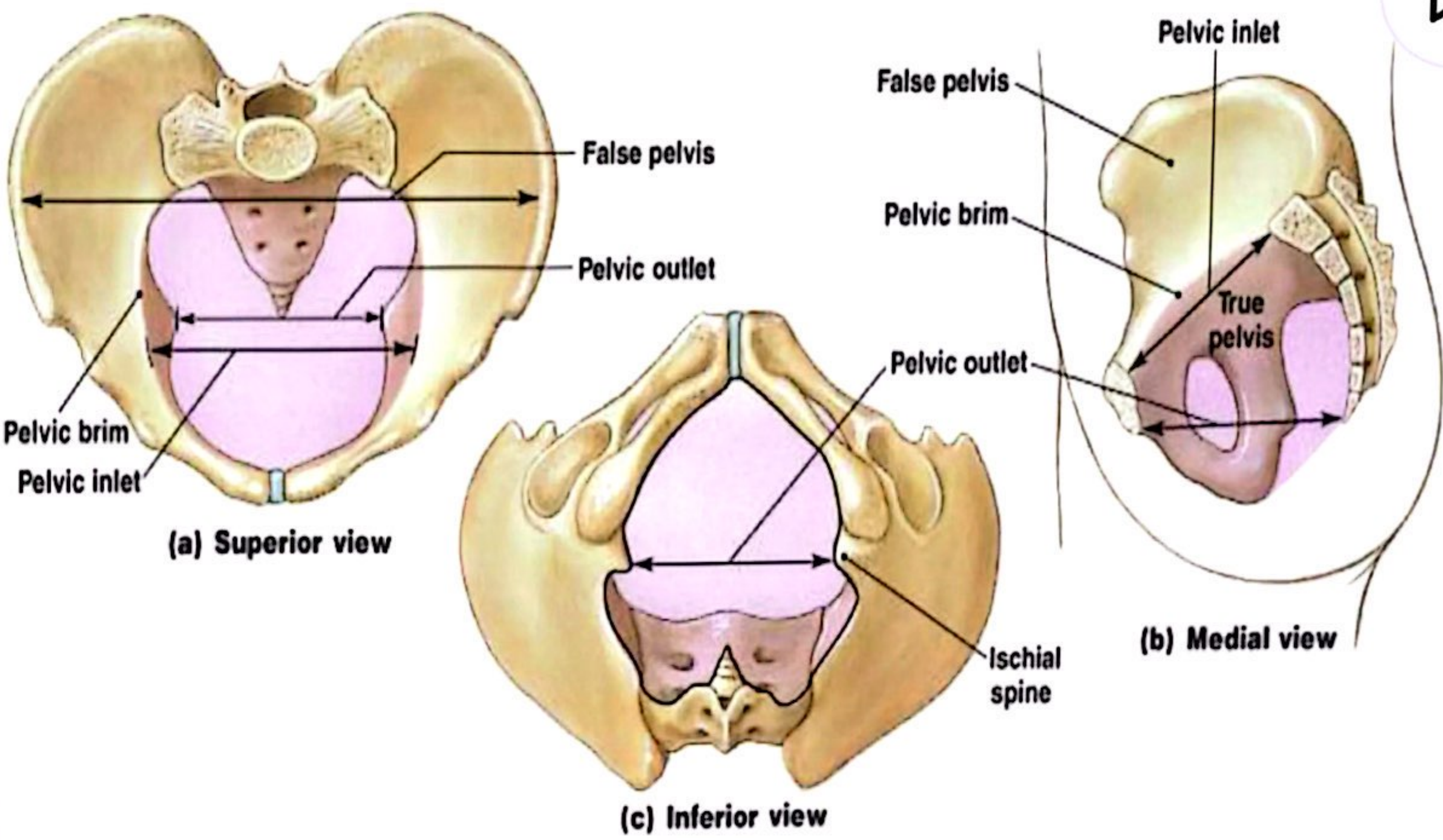


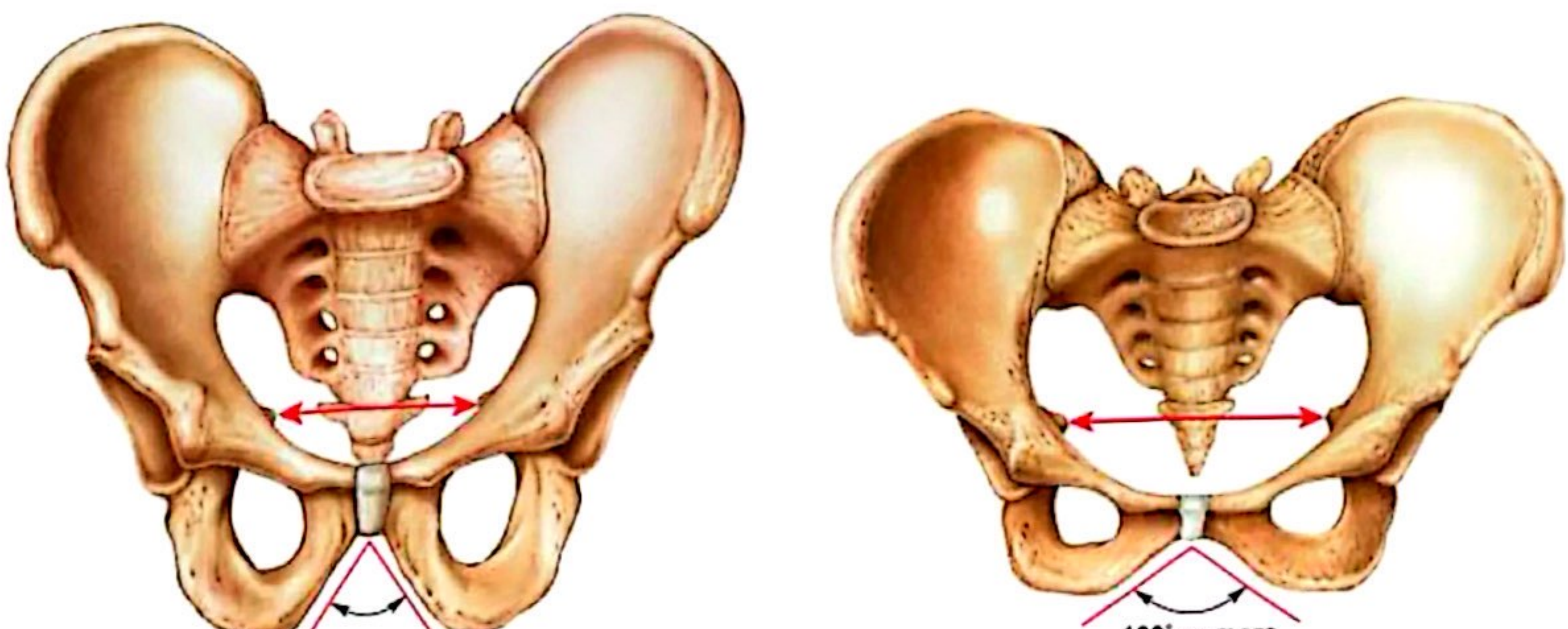
Figure 8-9

What are the structural and functional differences between the male and female pelvis?

Comparing the Male and Female Pelvis

- **Female pelvis:**
  - smoother
  - lighter
  - less prominent muscle and ligament attachments

Comparing the Male and Female Pelvis



Pelvis Modifications for Childbearing

- Enlarged pelvic outlet
- Broad pubic angle (> 100°)
- Less curvature of sacrum and coccyx
- Wide, circular pelvic inlet
- Broad, low pelvis
- Ilia project laterally, not upwards

What are the bones of the lower limbs, their functions, and features?

The Lower Limbs

- Functions:
  - weight bearing
  - motion

Note: leg = lower leg; thigh = upper leg

Bones of the Lower Limbs

- Femur (thigh)
- Patella (kneecap)
- Tibia and fibula (leg)
- Tarsals (ankle)
- Metatarsals (foot)
- Phalanges (toes)

The Femur (longest, heaviest)

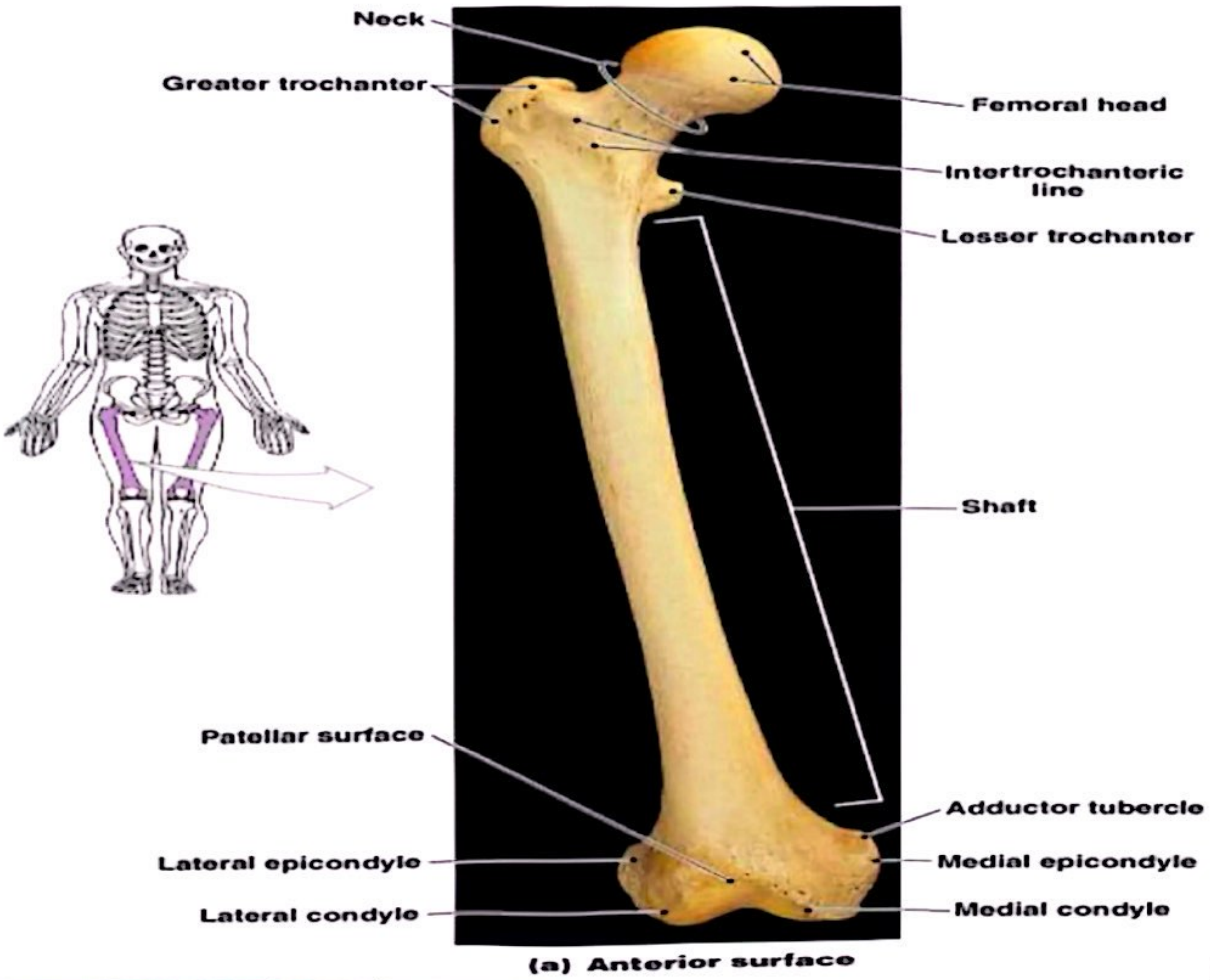
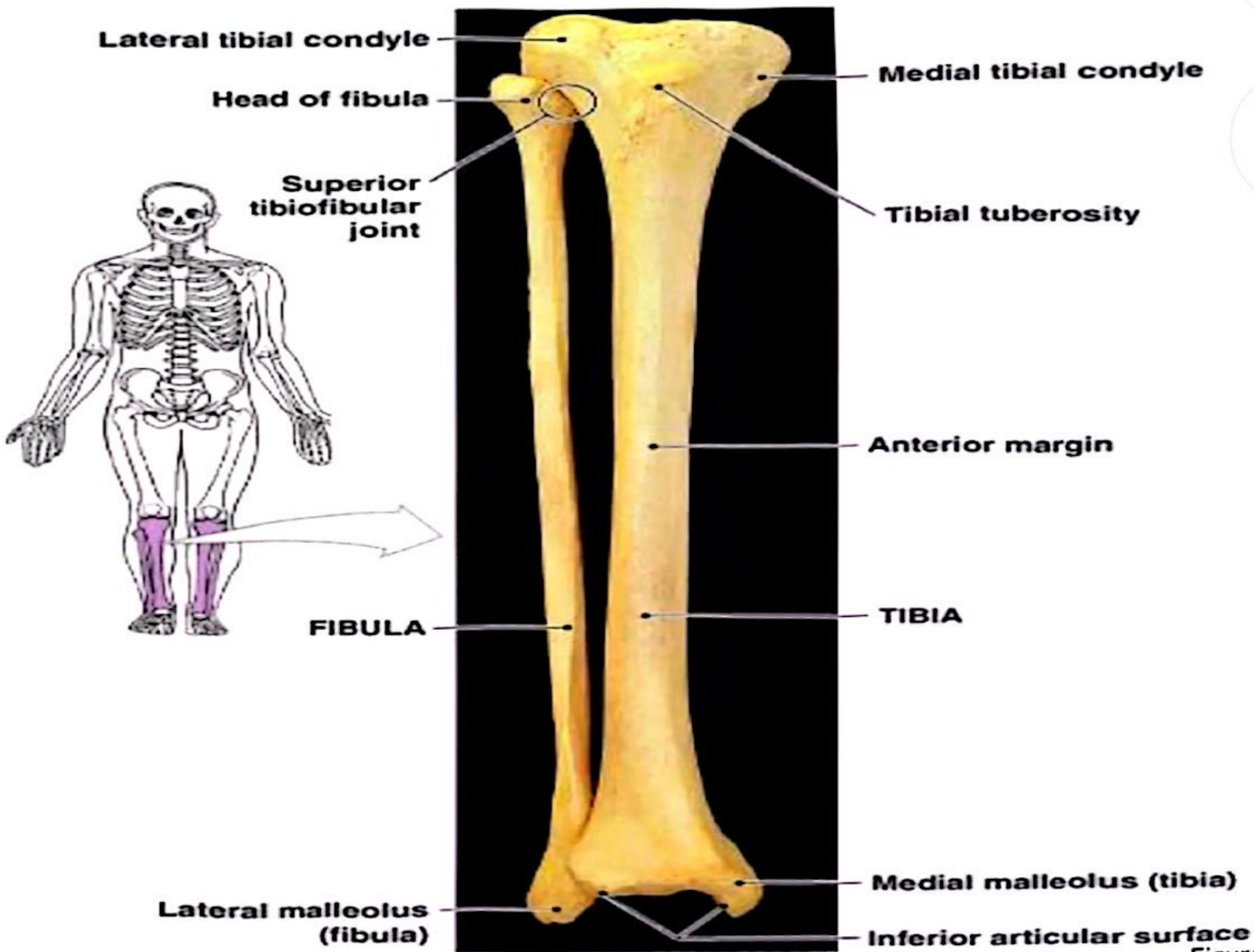


Figure 8-11

The Patella

- Also called the *kneecap*
- Formed within tendon of *quadriceps femoris*

The Tibia and Fibula



(a) Anterior view

Figure 8-13

The Tibia

- Also called the *shinbone*
- Supports body weight
- Larger than fibula
- Medial to fibula

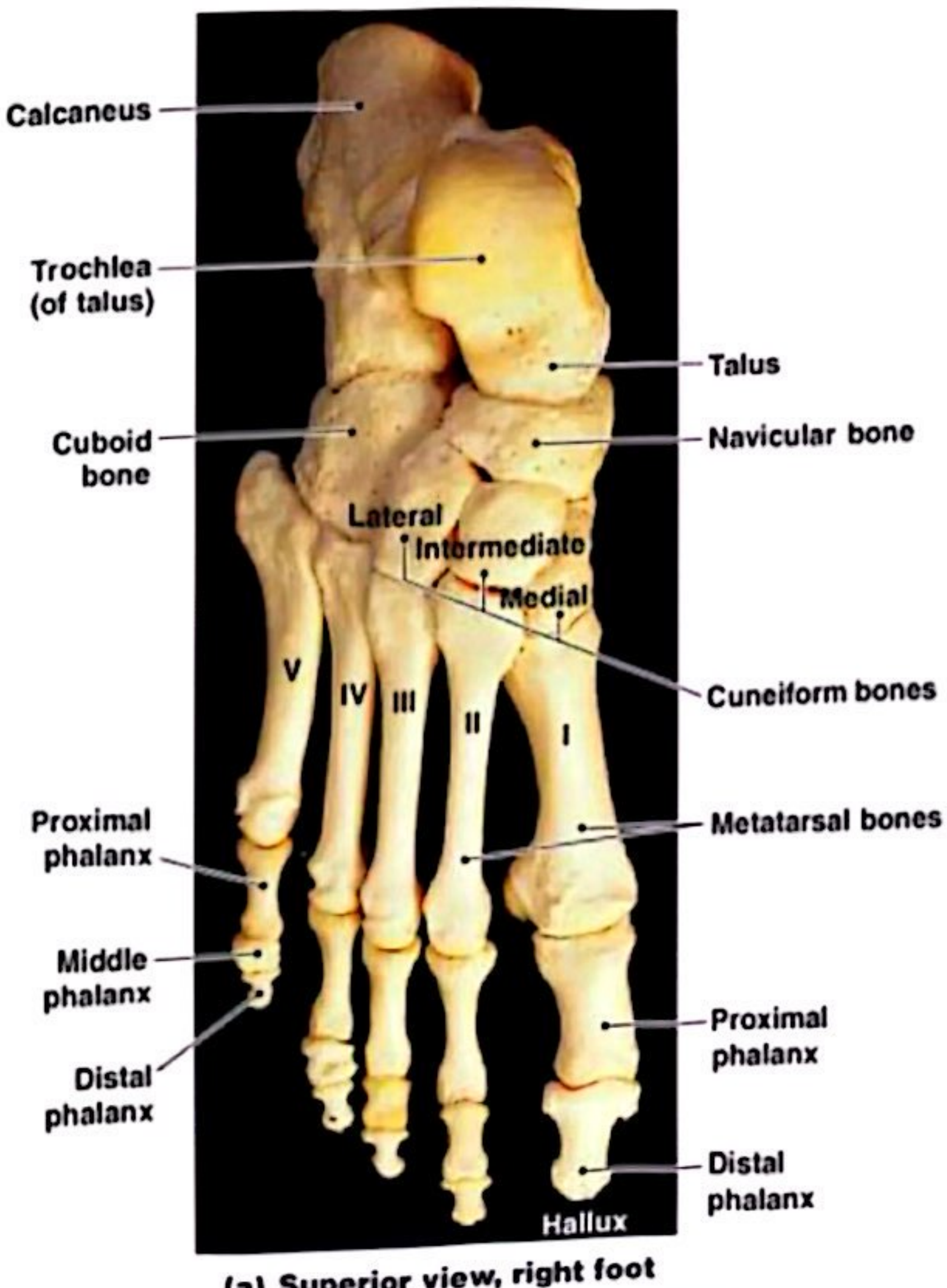
The Fibula

- Attaches muscles of feet and toes
- Smaller than tibia
- Lateral to tibia

Bones of the Ankle (Tarsals)

- Talus:
- Calcaneus (heel bone):
  - transfers weight to ground
  - attaches Achilles tendon

The Ankle



(a) Superior view, right foot

- Also called the *tarsus:*
  - consists of 7 tarsal bones

Figure 8-14a

Figure 8-14a

Feet: Metatarsal Bones

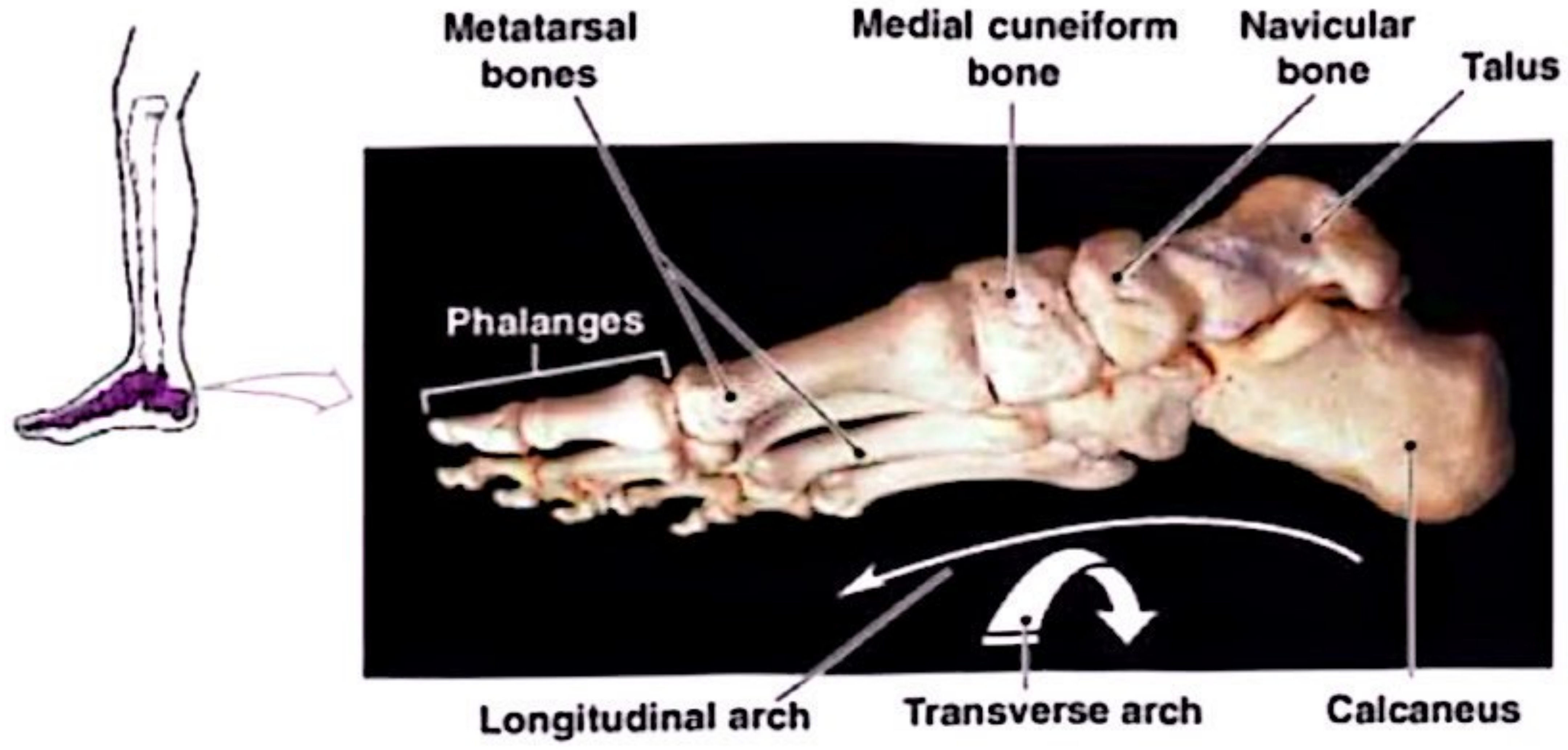
- 5 long bones of foot
- Numbered I–V, medial to lateral
- Articulate with toes

Toes: Phalanges

- Phalanges:
  - bones of the toes
- Hallux:
  - *big toe*, 2 phalanges (distal, proximal)
- Other 4 toes:
  - 3 phalanges (distal, medial, proximal)

Feet: Arches

- Arches transfer weight from 1 part of the foot to another



(b) Medial view, right foot

Figure 8-14b

Arches of the Foot

- Bones are arranged to form THREE strong arches, 2 longitudinal (medial & lateral)& 1 transverse
- Ligaments & tendons help to hold the bones firmly in the arched position but still allow a certain amount of spriginess
- Week arches are referred to as flat foot

Articulations (Joints)

Function

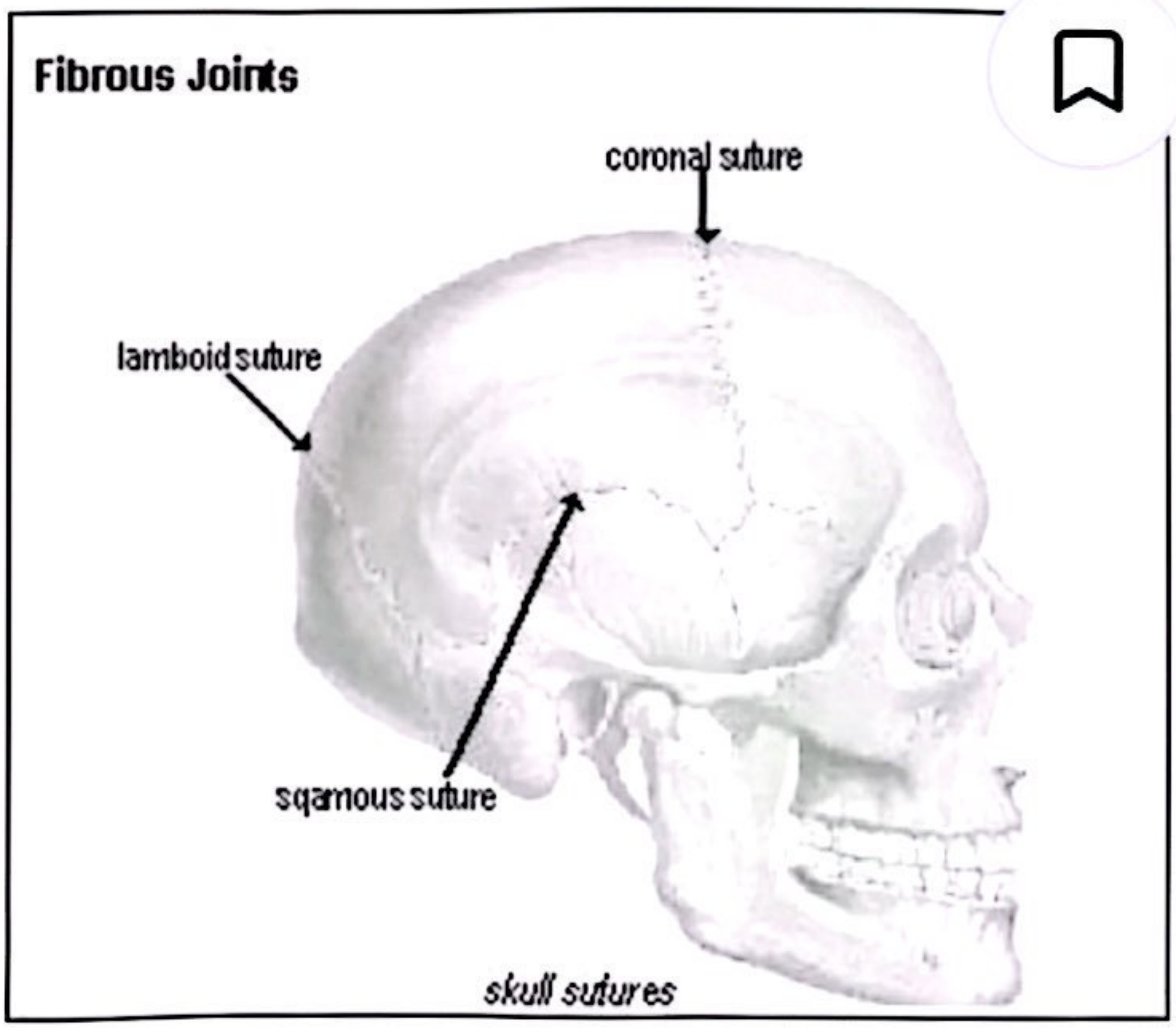
- Holds bones together
- Allows bones to move
- All bones articulate with at least one other bone except the hyoid.

Classification of joints

- Functional classification: focuses on the amount of movement (synarthrosis, amphiarthrosis and diarthrosis)
- Structural classification:based on whether Fibrous, Cartilage or a joint cavity separates the bony regions at the joint.
- As a general rule, fibrous joints are immovable and synovial joints are freely movable .

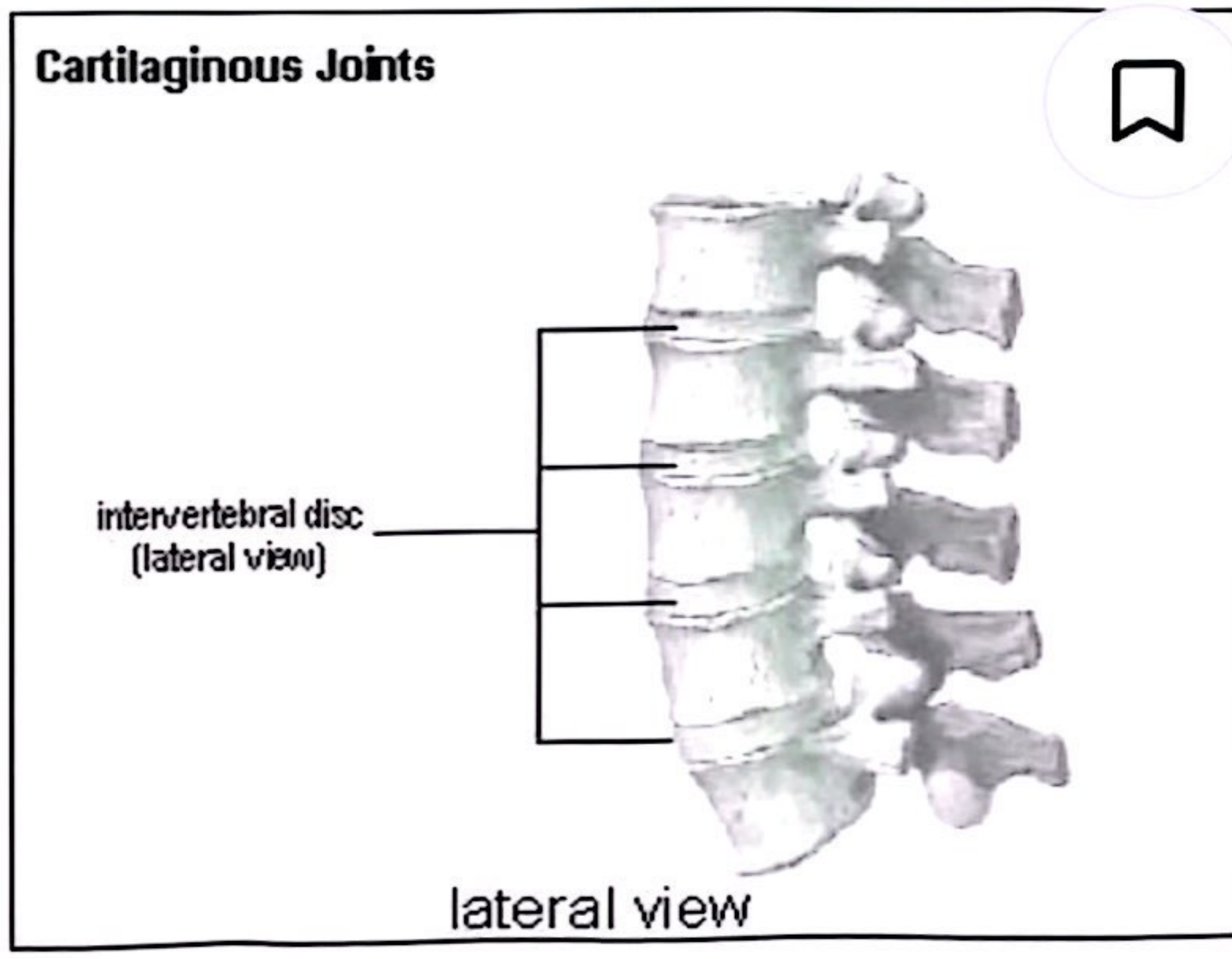
Types

- Synarthroses
- No movements
  - Primarily axial skeleton
  - Bones connected with fibrous tissue ligament
  - Examples: Skull sutures and distal Tibia/Fibula



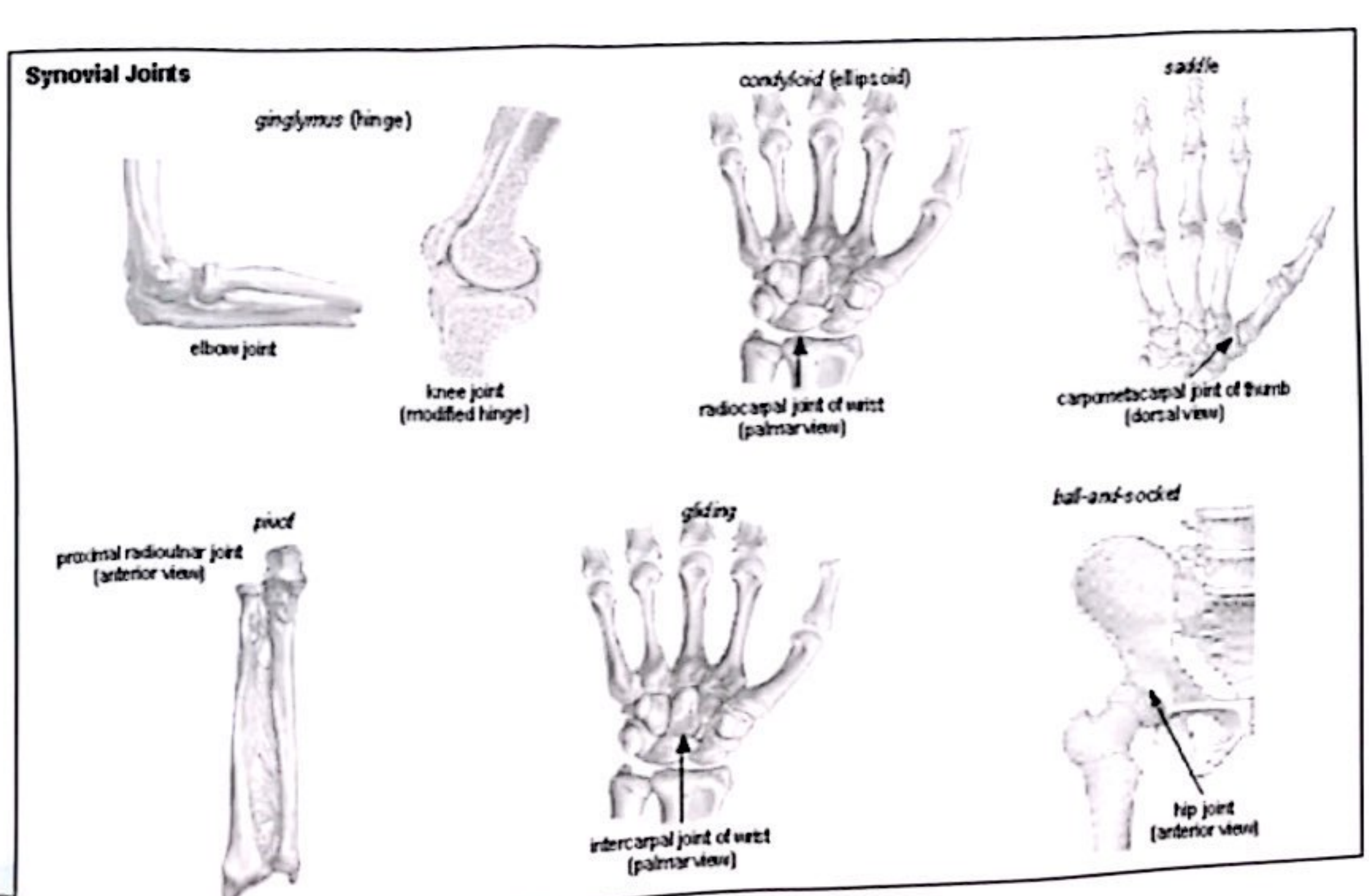
Types

- Amphiarthroses
  - Slightly movable
  - Axial skeleton
  - Connected by cartilage
  - Intervertebral joints, pubic symphysis



Types

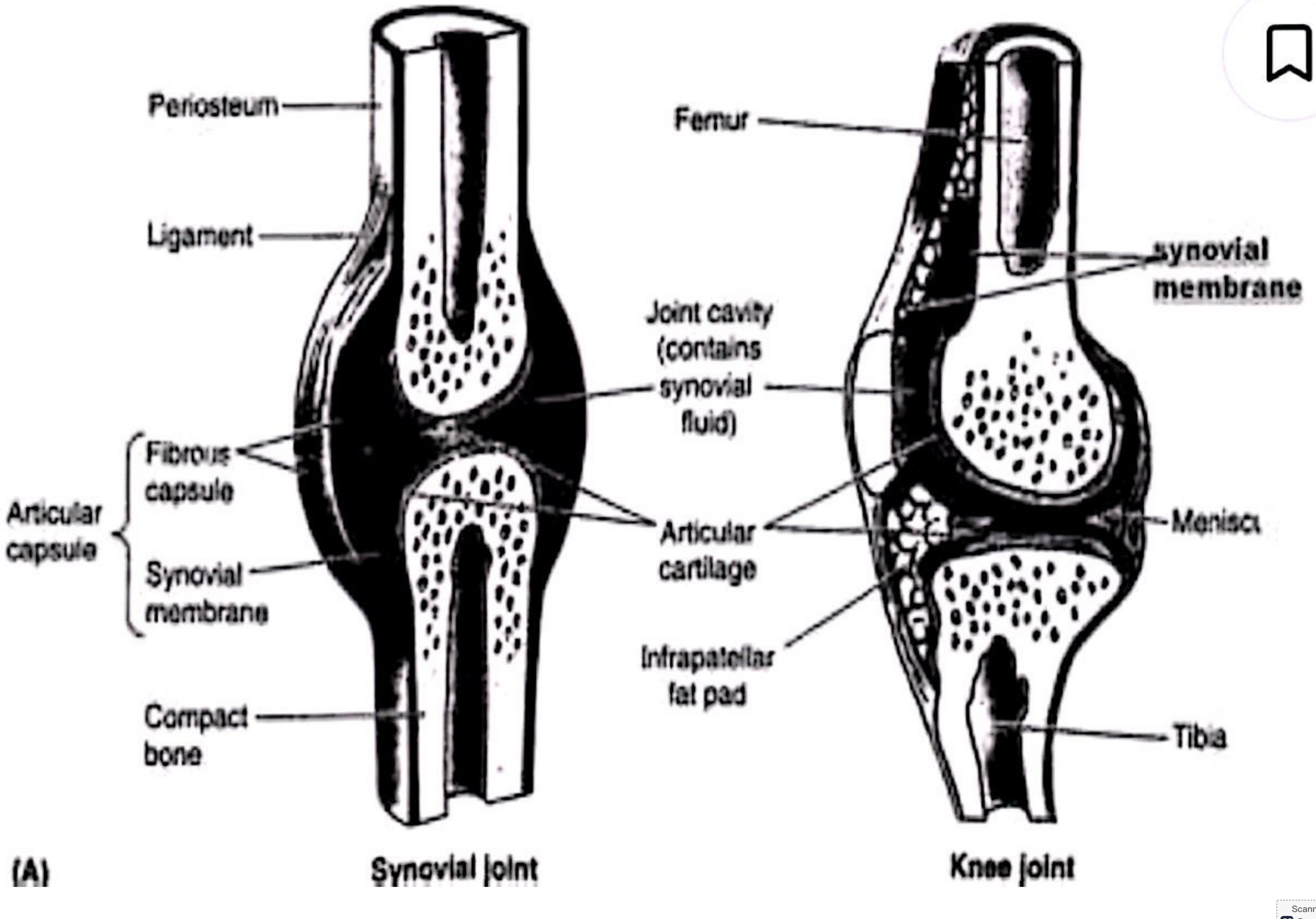
- Diarthroses – freely movable
  - Also called synovial (fluid filled joint cavity)
  - Primarily found in the limbs
  - Plane of movement depends on the joint

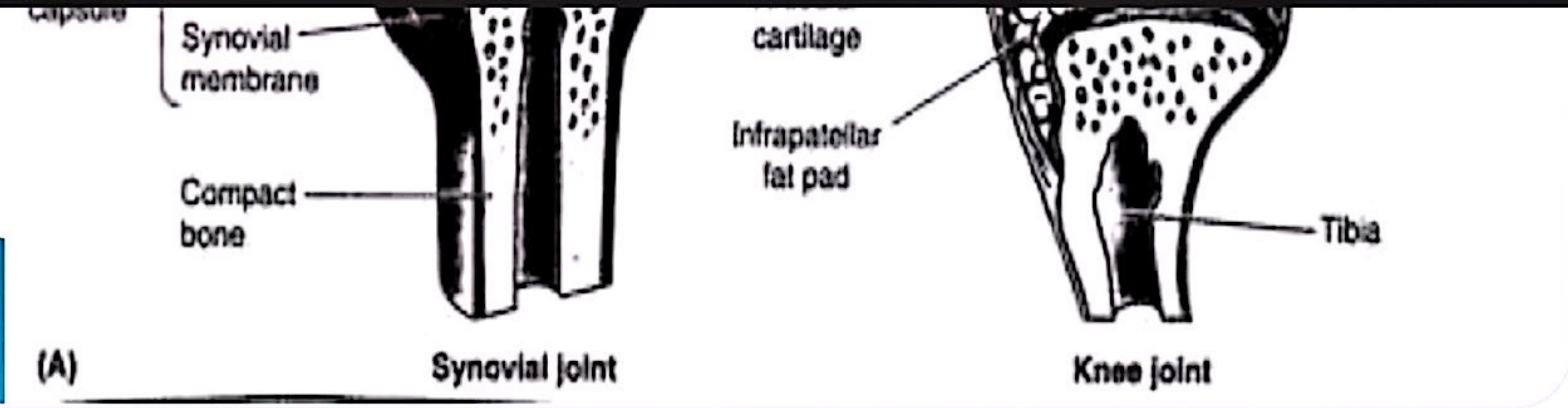


Synovial Joints – Structure

1. Articular cartilage: hyaline
2. Joint Cavity: space filled with lubricating fluid
3. Fibrous Capsule: fibrous CT lined with a smooth synovial membrane
4. Reinforcing Ligament: can be inside or outside the joint capsule
5. Synovial Fluid: viscous and lubricating
6. Tendons sheath an elongated bursa that rapes around a tendon subjected to friction.
7. Menisci: cartilaginous discs

Synovial Joint





Disorders of joints

- **Dislocation**: Bone is forced out of its position, Reduction is done by experts only
- **Sprain**: excessive stretch on a ligament
- **Arthritis**: inflammation of joints, may be
  - –Acute: usually bacterial
  - –Chronic: Rheumatoid ,Osteoarthritis and Gouty arthritis